

Fuzzy Inference Systems

CSC3034 Computational Intelligence

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Fuzzy Rules

Fuzzy Inference

Mamdani-Style Inference

Takagi-Sugeno-Kang-Style Inference

Building A Fuzzy Expert System

What is a fuzzy rule?

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IF x is A
THEN y is B

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THEN y is B

x, y : linguistic variables

A, B : linguistic values

Classical rules

A classical rule uses binary logic.

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Rule 1

IF speed is > 100

THEN stopping_distance is long

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Rule 1

IF speed is > 100

THEN stopping_distance is long

Rule 2

IF speed is < 400

THEN stopping_distance is short

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IF speed is < 400

THEN stopping_distance is short

- The variable 'speed' can have any numerical value.

A classical rule uses binary logic.

Rule 1

IF speed is > 100
THEN stopping_distance is long

Rule 2

IF speed is < 400
THEN stopping_distance is short

- ▶ The variable 'speed' can have any numerical value.
- ▶ But the linguistic variable 'stopping_distance' can only take the value of either 'long' or 'short'.

Fuzzy rules

Using the same variables,

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Rule 1

IF speed is fast

THEN stopping_distance is long

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Rule 1

IF speed is fast

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Rule 2

IF speed is slow

THEN stopping_distance is short

Using the same variables,

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- The linguistic variable 'speed' can also take any numerical value (translated to the corresponding fuzzy sets: 'slow', 'medium', and 'fast').

Using the same variables,

Rule 1

IF speed is fast

THEN stopping_distance is long

Rule 2

IF speed is slow

THEN stopping_distance is short

- ▶ The linguistic variable 'speed' can also take any numerical value (translated to the corresponding fuzzy sets: 'slow', 'medium', and 'fast').
- ▶ The linguistic variable 'stopping_distance' can be numerical value (translated from the corresponding fuzzy sets: 'short', 'medium', and 'long').

Classical rule-based systems:

Rule 1

IF speed is > 100

THEN stopping_distance is long

Classical rule-based systems:

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IF speed is > 100 $\rightarrow$ antecedent

THEN stopping_distance is long

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Classical rule-based systems:

Rule 1

IF speed is > 100 $\rightarrow$ antecedent

THEN stopping_distance is long $\rightarrow$ consequent

If the rule antecedent is **true**, then the consequent is **true**.

Fuzzy systems:

Rule 1

IF speed is fast

THEN stopping_distance is long

Fuzzy reasoning

Fuzzy systems:

Rule 1

IF speed is fast $\rightarrow$ antecedent

THEN stopping_distance is long

Fuzzy systems:

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Fuzzy systems:

Rule 1

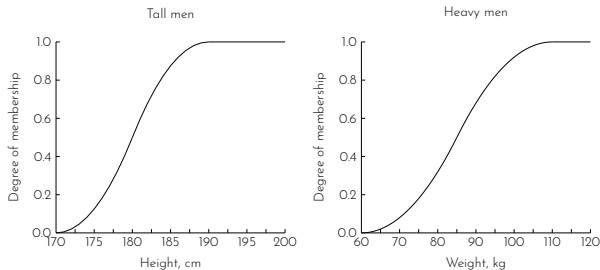
IF speed is fast → antecedent

THEN stopping_distance is long $\rightarrow$ consequent

If the rule antecedent is **true to some degree of membership**, then the consequent is **true to the same degree**.

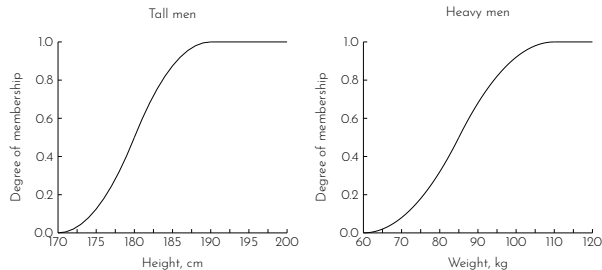
Example 1

Consider two fuzzy sets 'tall men' and 'heavy men'.



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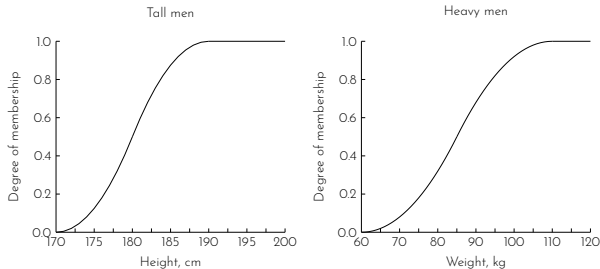
Given the relationship between a man's height and his weight,

Rule 1

IF height is tall
THEN weight is heavy

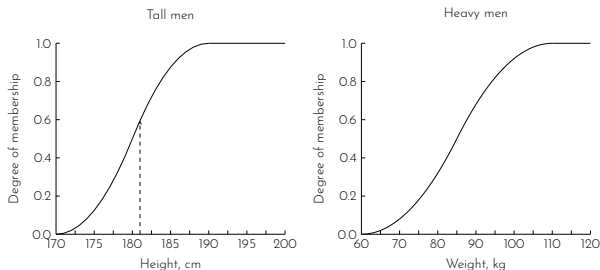
Example 1

For a man with height 181 cm, his weight can be derived from the fuzzy sets.



Example 1

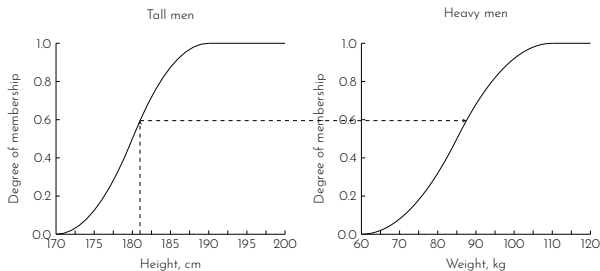
For a man with height 181 cm, his weight can be derived from the fuzzy sets.



$$181 \text{ cm} \rightarrow 0.595$$

Example 1

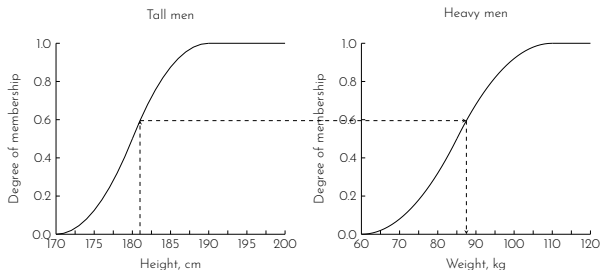
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For a man with height 181 cm, his weight can be derived from the fuzzy sets.



$$181 \text{ cm} \rightarrow 0.595 \rightarrow 87.5 \text{ kg}$$

Fuzzy rule with multiple parts of antecedent:

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IF project_duration is long
AND project_staffing is large
AND project_funding is inadequate
THEN risk is high

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All parts of the antecedent are calculated and resolved to a single number using fuzzy set operations.

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$$A \text{ AND } B = \min(\mu_A, \mu_B)$$

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All parts of the antecedent are calculated and resolved to a single number using fuzzy set operations.

$$A \text{ AND } B = \min(\mu_A, \mu_B)$$

$$A \text{ OR } B = \max(\mu_A, \mu_B)$$

Fuzzy rule with multiple parts of consequent:

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IF temperature is hot
THEN hot_water is reduced;
 cold_water is increased

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THEN hot_water is reduced;
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All parts of the consequent are affected equally by the antecedent.

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Step 1: Fuzzification of input variables

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Step 2: Rule evaluation

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- Step 3: Aggregation of rule consequents

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- Step 4: Defuzzification

Fuzzy inference

There are two different techniques for fuzzy inference.

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Mamdani-style inference

Takagi-Sugeno-Kang-style inference

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Mamdani-style inference

The rule consequent is a fuzzy set.

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Mamdani-style inference

The rule consequent is a fuzzy set.

IF X_1 is A_1 and ... and X_n is A_n
THEN Y is B_i

Takagi-Sugeno-Kang-style inference

There are two different techniques for fuzzy inference.

Mamdani-style inference

The rule consequent is a fuzzy set.

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Takagi-Sugeno-Kang-style inference

The rule consequent is a mathematical function.

There are two different techniques for fuzzy inference.

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The rule consequent is a fuzzy set.

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Takagi-Sugeno-Kang-style inference

The rule consequent is a mathematical function.

IF X_1 is A_1 and ...and X_n is A_n
THEN $Y = p_0 + p_1X_1 + \dots + p_nX_n$

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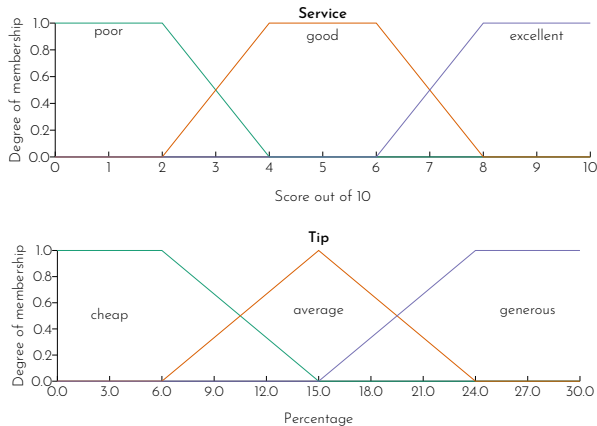
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THEN $Y = p_0 + p_1X_1 + \dots + p_nX_n$

Mamdani-style is the more commonly used method.

Mamdani-style inference

Given a fuzzy system with two linguistic variables 'service' and 'tip' with the following fuzzy sets.



The following Mamdani-style rule applies to the system.

IF service is good
THEN tip is average

If a customer then rates the service at the restaurant as being a 3 out of 10, what would be the percentage of the given tip?

Mamdani-style inference

Step 1: Fuzzification

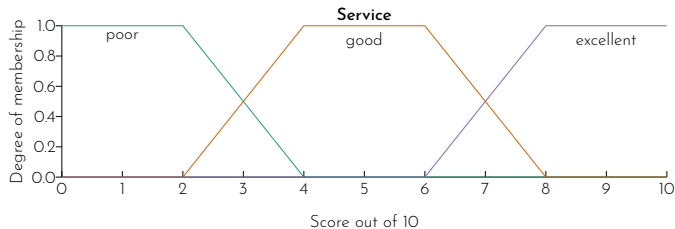
Mamdani-style inference

Step 1: Fuzzification

Identify the degree of membership of the input variable in each fuzzy set.

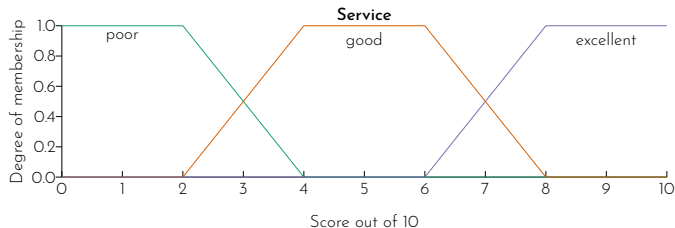
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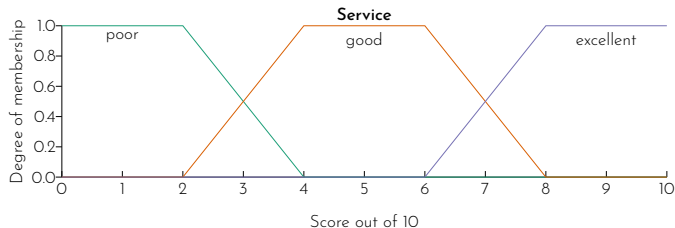
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Let X denotes the linguistic variable 'service'.

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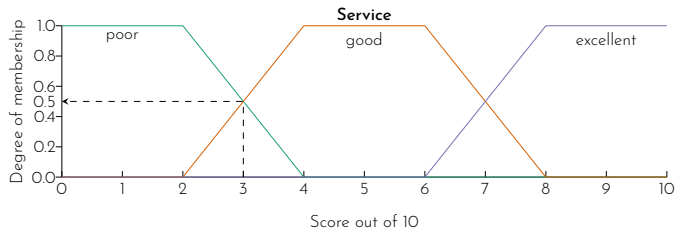
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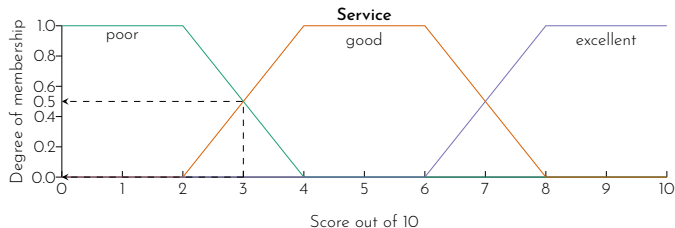
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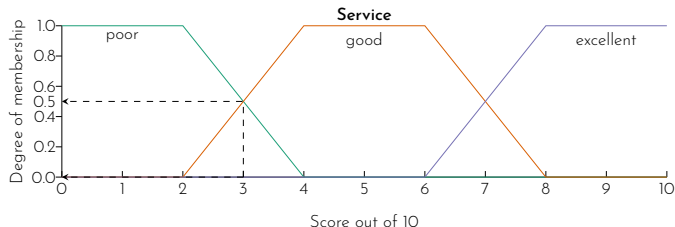
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Step 1: Fuzzification

Identify the degree of membership of the input variable in each fuzzy set.



Let X denotes the linguistic variable 'service'. For the service with score 3 out of 10,

$$\mu_{X=\text{poor}} = 0.5$$

$$\mu_{X=\text{good}} = 0.5$$

$$\mu_{X=\text{excellence}} = 0.0$$

Mamdani-style inference

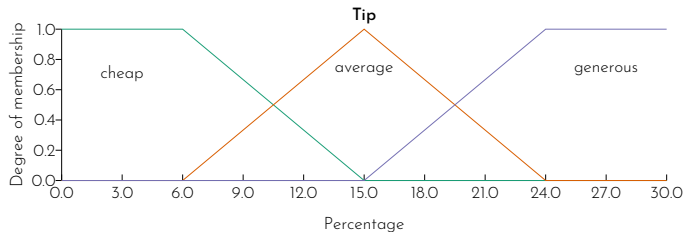
Step 2: Rule evaluation

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Fire the rule to the same degree.

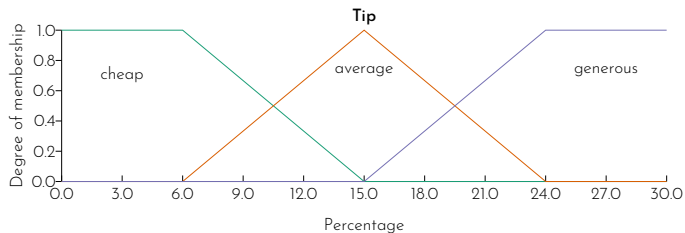
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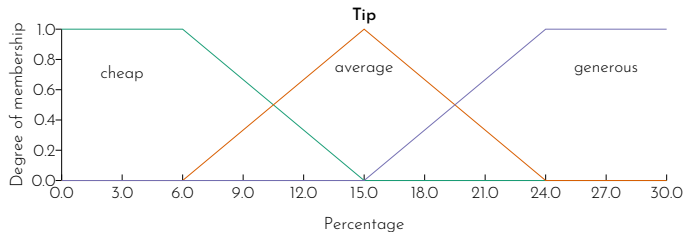
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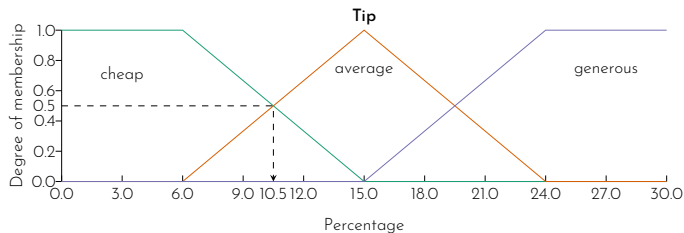
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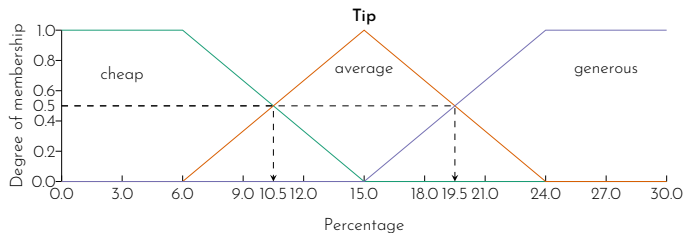
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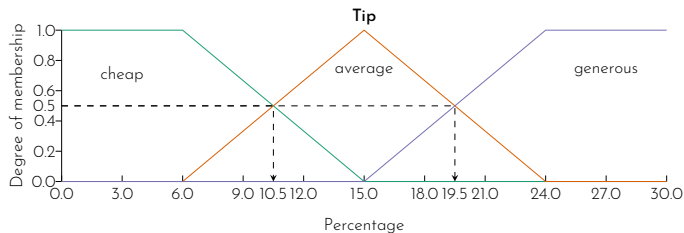
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THEN tip is average

The tip amount in 'average' is between 10.5% and 19.5%.

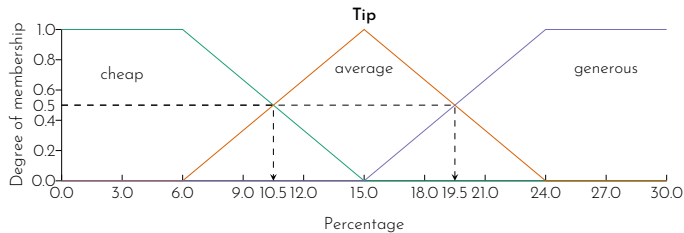
Step 2: Rule evaluation (continued)

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The membership function of the consequent is then 'clipped' or 'scaled'.

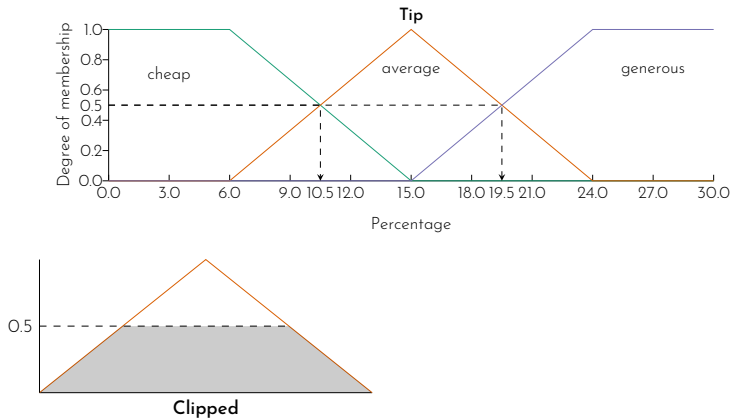
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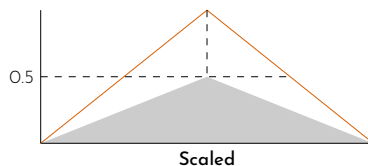
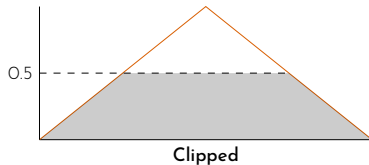
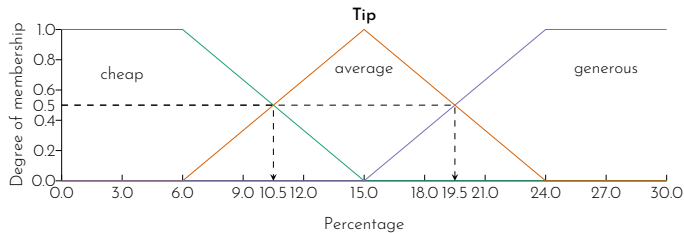
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- ▶ This step is important if multiple rules are fired.
- ▶ We will look at this in the latter example.

Step 4: Defuzzification

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- Identify the crisp number to be the final output of the fuzzy system.

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- ▶ The centroid technique is commonly used to identify the crisp output.

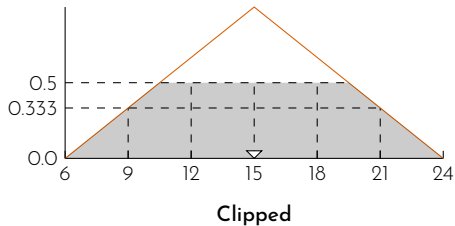
Step 4: Defuzzification

- ▶ Identify the crisp number to be the final output of the fuzzy system.
- ▶ The centroid technique is commonly used to identify the crisp output.
- ▶ The centroid (centre of gravity) is identified using

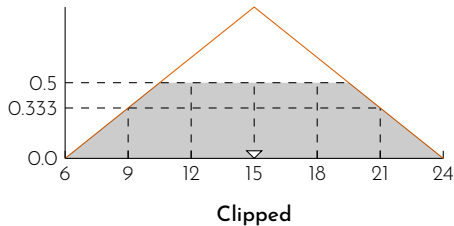
$$\text{centre of gravity} = \frac{\sum_{x=a}^b \mu_A \cdot x}{\sum_{x=a}^b \mu_A}$$

Step 4: Defuzzification (continued)

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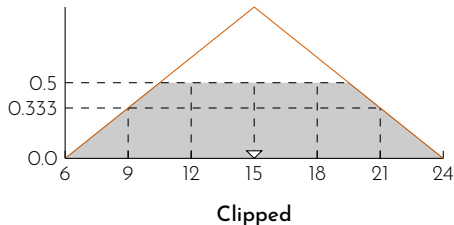


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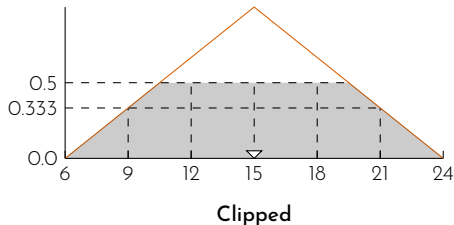
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Step 4: Defuzzification (continued)



$$\text{centre of gravity} = \frac{\sum_{x=a}^b \mu_A \cdot x}{\sum_{x=a}^b \mu_A} = \frac{(6 + 24)(0) + (9 + 21)(0.333) + (12 + 15 + 18)(0.5)}{2(0) + 2(0.333) + 3(0.5)}$$

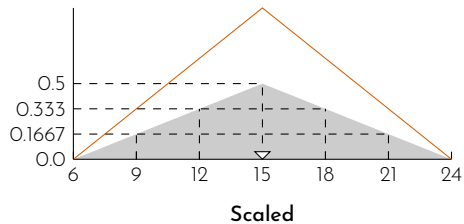
Step 4: Defuzzification (continued)



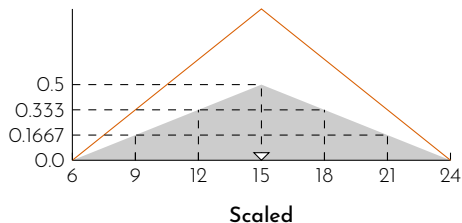
$$\begin{aligned}\text{centre of gravity} &= \frac{\sum_{x=a}^b \mu_A \cdot x}{\sum_{x=a}^b \mu_A} = \frac{(6 + 24)(0) + (9 + 21)(0.333) + (12 + 15 + 18)(0.5)}{2(0) + 2(0.333) + 3(0.5)} \\ &= 15\end{aligned}$$

Step 4: Defuzzification (continued)

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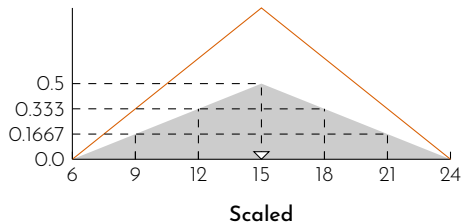


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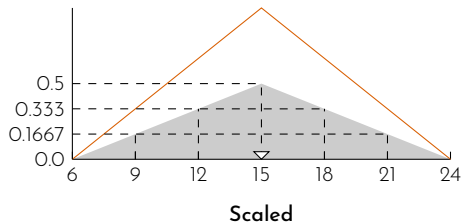
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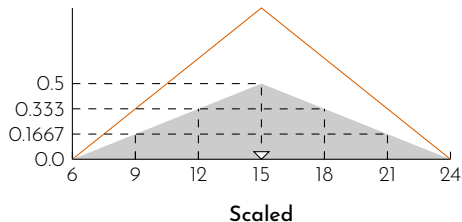
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The final output of the fuzzy system, i.e. the percentage of the tip that would be given is 15%.

Takagi-Sugeno-Kang-style inference

Sugeno-style inference shortens the time of fuzzy inference by using a singleton as the membership function of the rule consequent.

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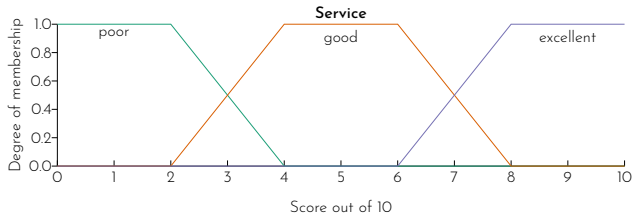
The previous example is modified slightly for Sugeno-style inference.

Sugeno-style inference

We have the linguistic variable 'service' with the same fuzzy sets

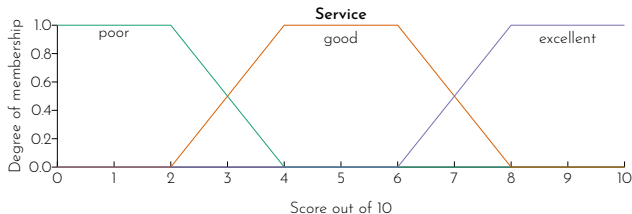
Sugeno-style inference

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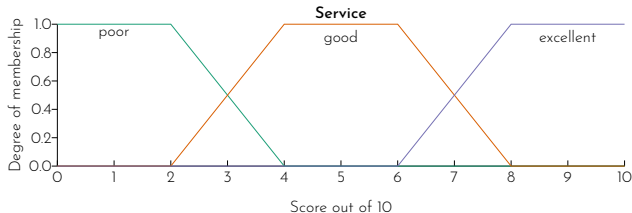


with the Sugeno-style rule

IF service is good
THEN tip is $3 \times \text{service} \%$

Sugeno-style inference

We have the linguistic variable 'service' with the same fuzzy sets



with the Sugeno-style rule

IF service is good
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If a customer then rates the service at the restaurant as being a 3 out of 10, what would be the percentage of the given tip?

Sugeno-style inference

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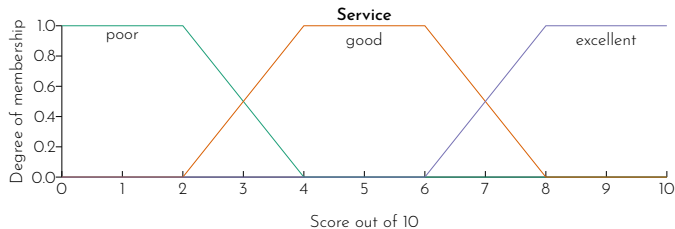
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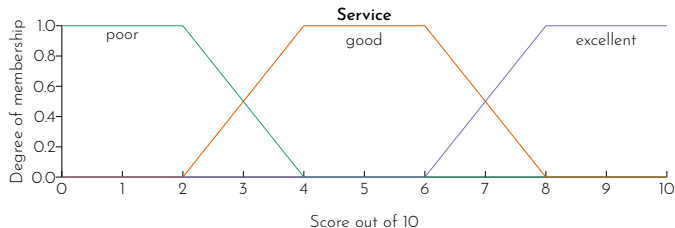
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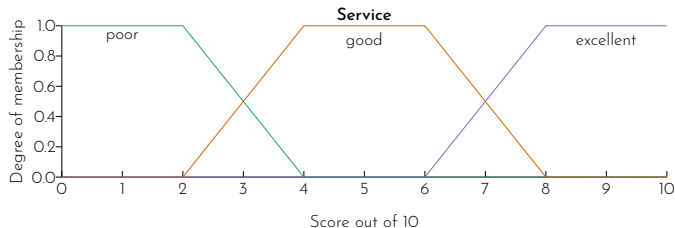
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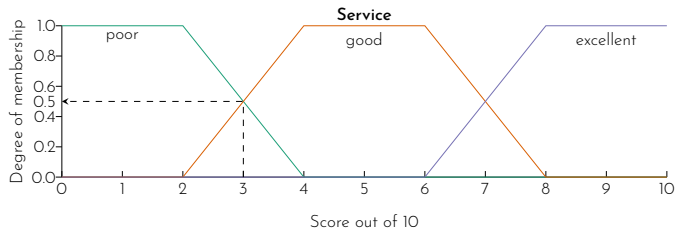
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Let X denotes the linguistic variable 'service'. For the service with score 3 out of 10,

Step 1: Fuzzification

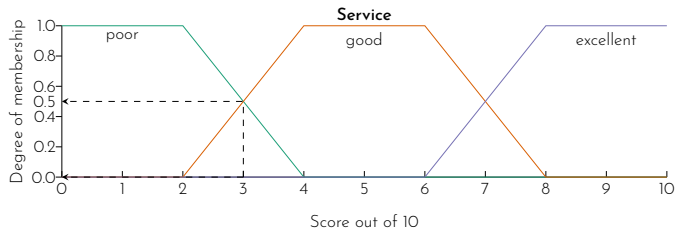
Identify the degree of membership of the input variable in each fuzzy set.



Let X denotes the linguistic variable 'service'. For the service with score 3 out of 10,

Step 1: Fuzzification

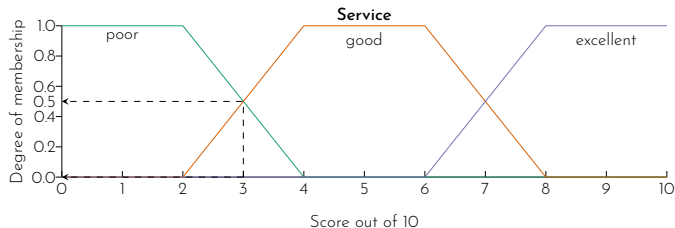
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Step 1: Fuzzification

Identify the degree of membership of the input variable in each fuzzy set.



Let X denotes the linguistic variable 'service'. For the service with score 3 out of 10,

$$\mu_{X=\text{poor}} = 0.5$$

$$\mu_{X=\text{good}} = 0.5$$

$$\mu_{X=\text{excellence}} = 0.0$$

Sugeno-style inference

Step 2: Rule evaluation

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According to this rule:

IF service is good $\rightarrow \mu_{X=\text{good}} = 0.5$
THEN tip is $3 \times \text{service} \%$

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THEN tip is $3 \times \text{service} \%$

$$\text{tip} = 3 \times 3 = 9\%$$

the firing strength of this rule is $\alpha = \mu_{X=\text{good}} = 0.5$

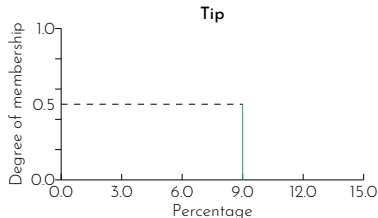
Step 2: Rule evaluation

According to this rule:

IF service is good $\rightarrow \mu_{X=\text{good}} = 0.5$
THEN tip is $3 \times \text{service} \%$

$$\text{tip} = 3 \times 3 = 9\%$$

the firing strength of this rule is $\alpha = \mu_{X=\text{good}} = 0.5$



Step 3: Aggregation of rule consequents

- ▶ Skip the step as there is only one rule fired.

Step 4: Defuzzification

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The final crisp output is obtained using the weighted average formula.

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$$\text{weighted average} = \frac{\sum_{x=a}^b \alpha_x \cdot x}{\sum_{x=a}^b \alpha_x}$$

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$$\text{weighted average} = \frac{\sum_{x=a}^b \alpha_x \cdot x}{\sum_{x=a}^b \alpha_x}$$

$$\therefore \text{tip} = \frac{0.5 \cdot 9}{0.5} = 9\%$$

Example 2

Consider the fuzzy system with two input variables x and y , and one output variable z . All the variables have the range between 0 and 10.

Variable	Fuzzy sets	Fit vector
x	A_1	$(1/3, 0/5)$
x	A_2	$(0/3, 1/7, 0/8)$
x	A_3	$(0/6, 1/8)$
y	B_1	$(1/3, 0/8)$
y	B_2	$(0/5, 1/8)$
z	C_1	$(1/2, 0/5)$
z	C_2	$(0/2, 1/4, 1/6, 0/8)$
z	C_3	$(0/7, 1/8)$

Example 2

The following Mamdani-style rules apply to this system.

Rule 1

IF x is A_3

OR y is B_1

THEN z is C_1

Rule 2

IF x is A_2

AND y is B_2

THEN z is C_2

Rule 3

IF x is A_1

THEN z is C_3

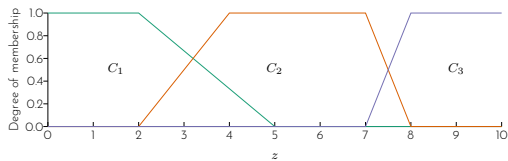
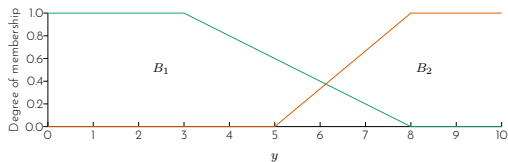
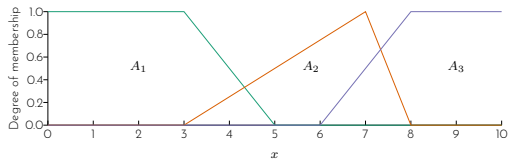
Given that x_1 is 3.5, y_1 is 6.5, what is the value of z_1 ?

Example 2

First, draw the membership function of the fuzzy sets on the graph paper.

Variable	Fuzzy sets	Fit vector
x	A_1	$(1/3, 0/5)$
x	A_2	$(0/3, 1/7, 0/8)$
x	A_3	$(0/6, 1/8)$
y	B_1	$(1/3, 0/8)$
y	B_2	$(0/5, 1/8)$
z	C_1	$(1/2, 0/5)$
z	C_2	$(0/2, 1/4, 1/6, 0/8)$
z	C_3	$(0/7, 1/8)$

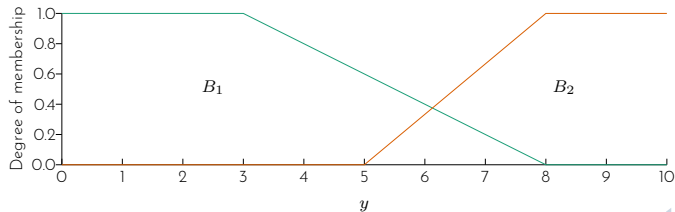
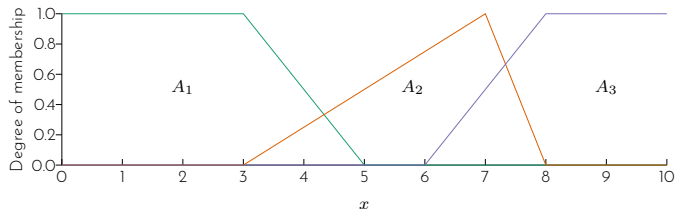
Example 2



Example 2

Step 1: Fuzzification

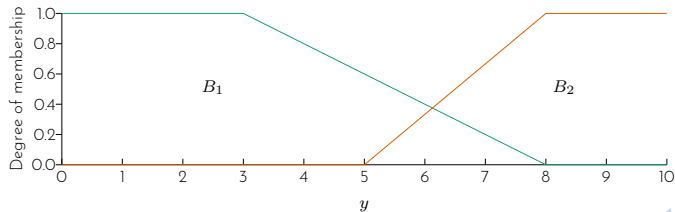
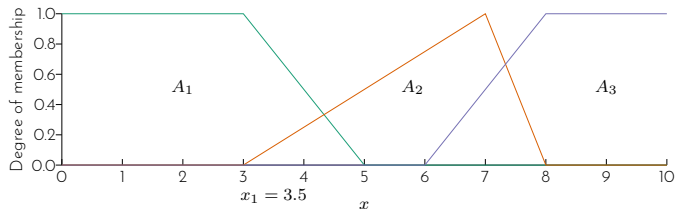
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 2

Step 1: Fuzzification

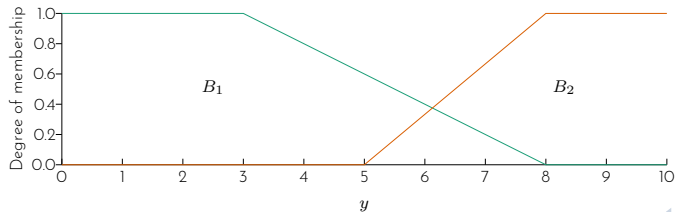
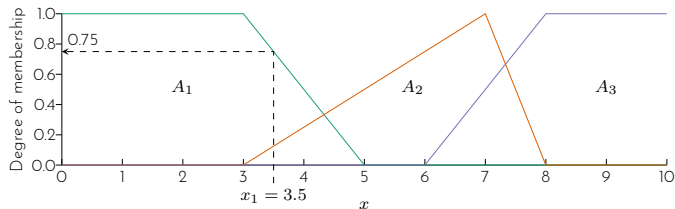
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Example 2

Step 1: Fuzzification

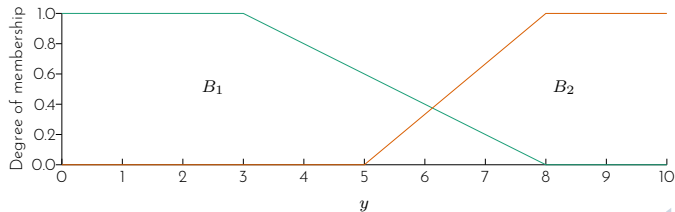
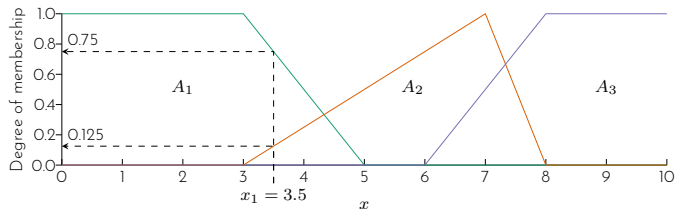
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 2

Step 1: Fuzzification

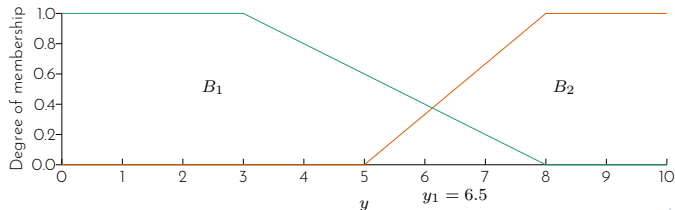
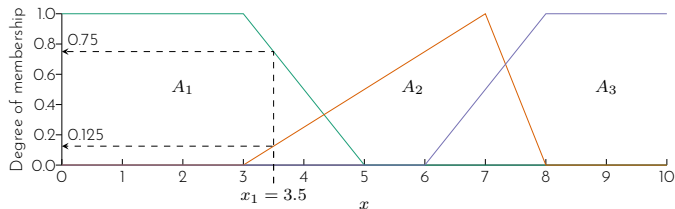
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 2

Step 1: Fuzzification

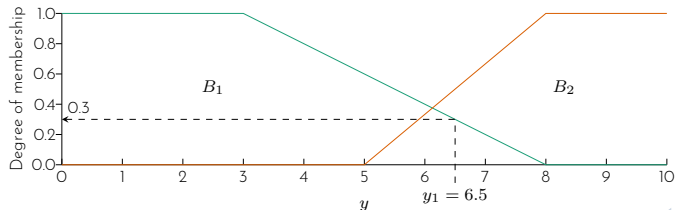
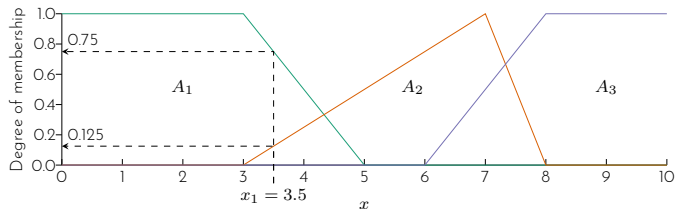
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 2

Step 1: Fuzzification

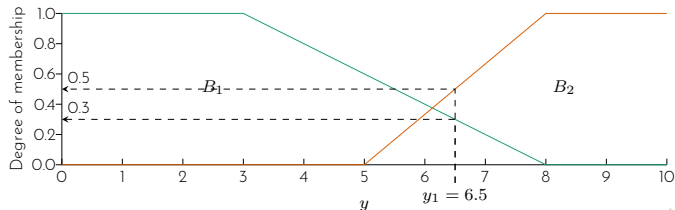
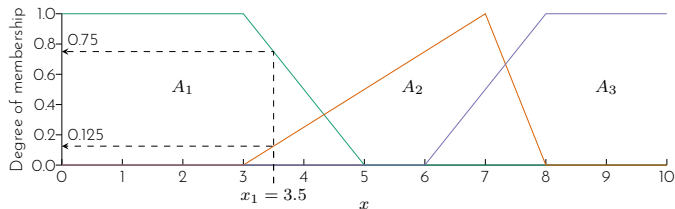
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 2

Step 1: Fuzzification

Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 2

Step 1: Fuzzification (continued)

Therefore we have

$$\mu_{x=A_1} = 0.75$$

$$\mu_{x=A_2} = 0.125$$

$$\mu_{x=A_3} = 0$$

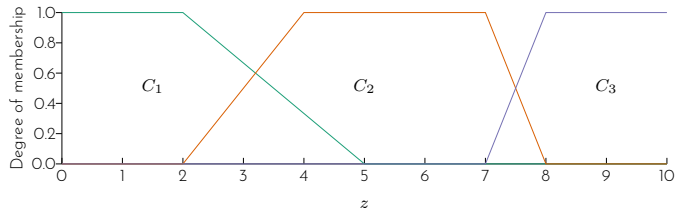
$$\mu_{y=B_1} = 0.3$$

$$\mu_{y=B_2} = 0.5$$

Example 2

Step 2: Rule evaluation

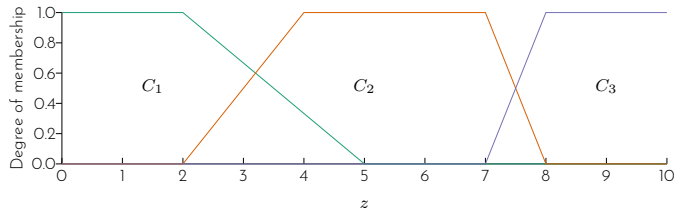
Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is C_1	



Example 2

Step 2: Rule evaluation

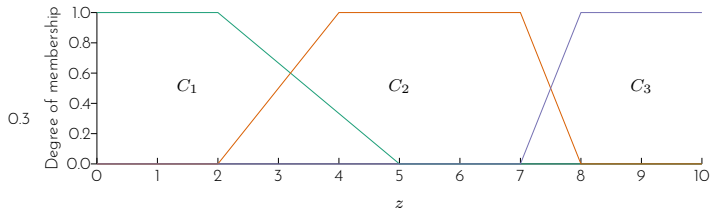
Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is C_1	$\max(0.0, 0.3) = 0.3$



Example 2

Step 2: Rule evaluation

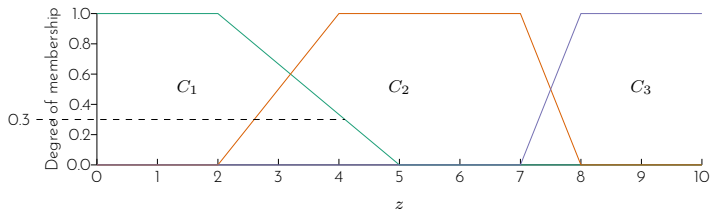
Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is C_1	$\max(0.0, 0.3) = 0.3$



Example 2

Step 2: Rule evaluation

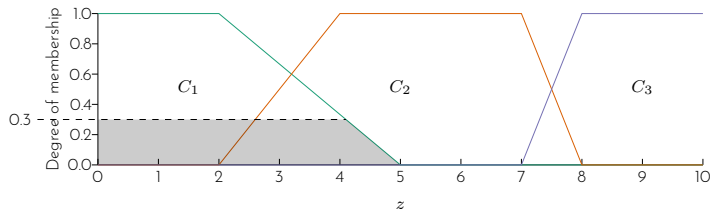
Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is C_1	$\max(0.0, 0.3) = 0.3$



Example 2

Step 2: Rule evaluation

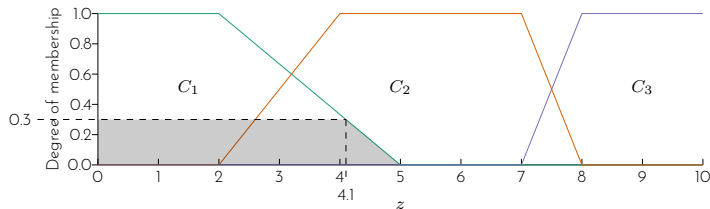
Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is C_1	$\max(0.0, 0.3) = 0.3$



Example 2

Step 2: Rule evaluation

Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is C_1	$\max(0.0, 0.3) = 0.3$



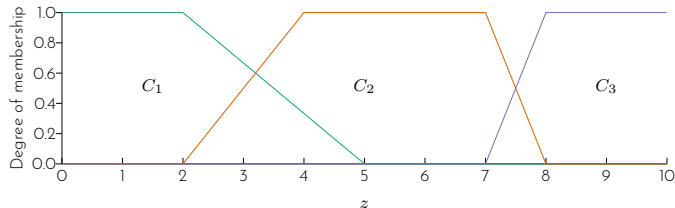
Example 2

Step 2: Rule evaluation (continued)

Rule 2

IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is C_2	

Degree of membership



Example 2

Step 2: Rule evaluation (continued)

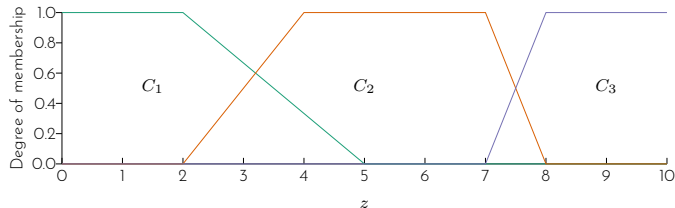
Rule 2

Degree of membership

IF x is A_2 0.125

AND y is B_2 0.5

THEN z is C_2 $\min(0.125, 0.5) = 0.125$



Example 2

Step 2: Rule evaluation (continued)

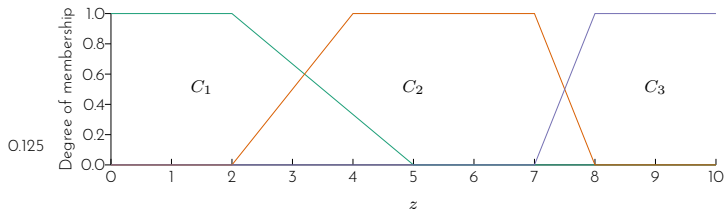
Rule 2

Degree of membership

IF x is A_2 0.125

AND y is B_2 0.5

THEN z is C_2 $\min(0.125, 0.5) = 0.125$



Example 2

Step 2: Rule evaluation (continued)

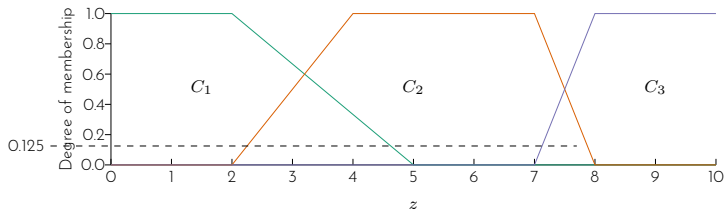
Rule 2

Degree of membership

IF x is A_2 0.125

AND y is B_2 0.5

THEN z is C_2 $\min(0.125, 0.5) = 0.125$



Example 2

Step 2: Rule evaluation (continued)

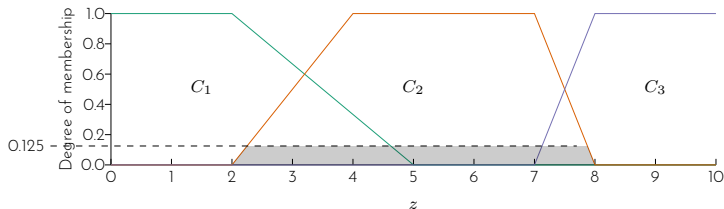
Rule 2

Degree of membership

IF x is A_2 0.125

AND y is B_2 0.5

THEN z is C_2 $\min(0.125, 0.5) = 0.125$



Example 2

Step 2: Rule evaluation (continued)

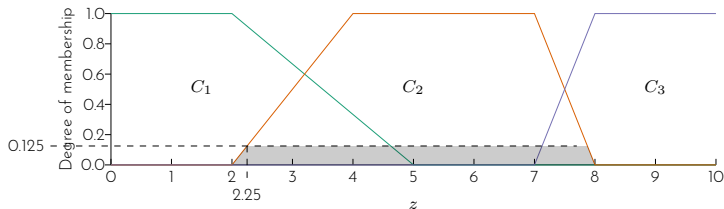
Rule 2

Degree of membership

IF x is A_2 0.125

AND y is B_2 0.5

THEN z is C_2 $\min(0.125, 0.5) = 0.125$



Example 2

Step 2: Rule evaluation (continued)

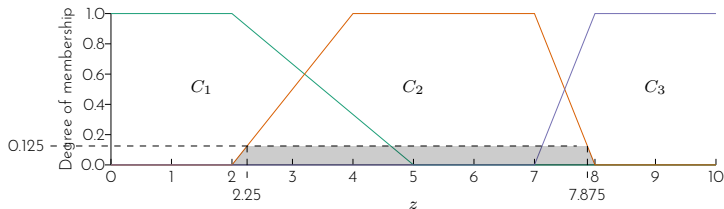
Rule 2

Degree of membership

IF x is A_2 0.125

AND y is B_2 0.5

THEN z is C_2 $\min(0.125, 0.5) = 0.125$



Example 2

Step 2: Rule evaluation (continued)

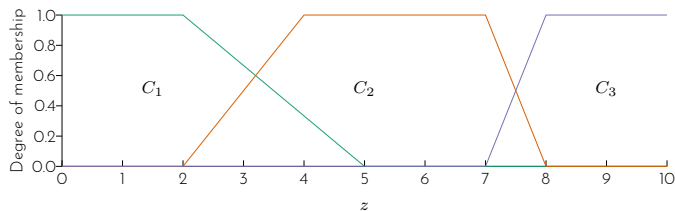
Rule 3

Degree of membership

IF x is A_1

0.75

THEN z is C_3



Example 2

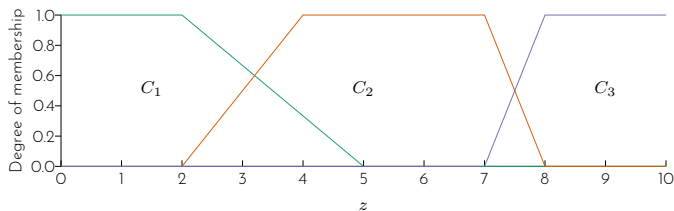
Step 2: Rule evaluation (continued)

Rule 3

IF x is A_1 0.75

THEN z is C_3 0.75

Degree of membership



Example 2

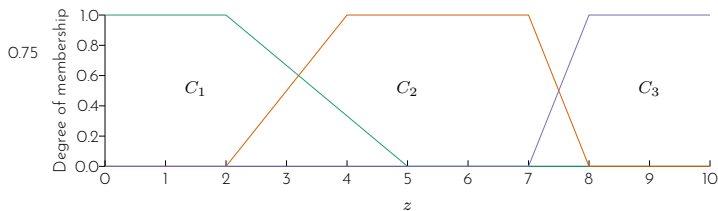
Step 2: Rule evaluation (continued)

Rule 3

IF x is A_1 0.75

THEN z is C_3 0.75

Degree of membership



Example 2

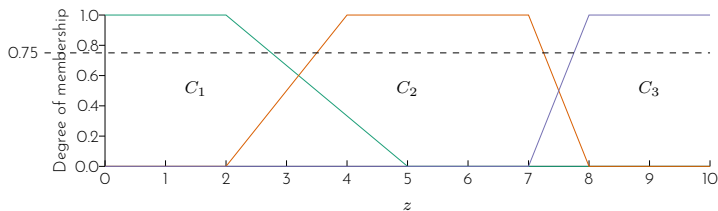
Step 2: Rule evaluation (continued)

Rule 3

IF x is A_1 0.75

THEN z is C_3 0.75

Degree of membership



Example 2

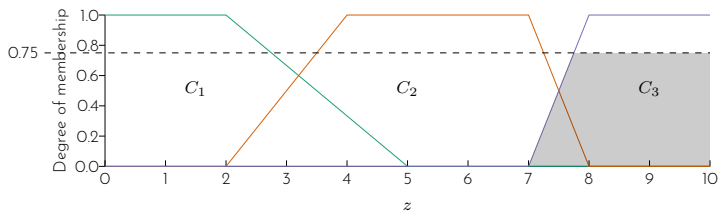
Step 2: Rule evaluation (continued)

Rule 3

IF x is A_1 0.75

THEN z is C_3 0.75

Degree of membership



Example 2

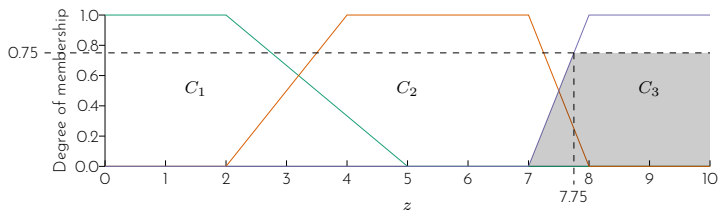
Step 2: Rule evaluation (continued)

Rule 3

IF x is A_1 0.75

THEN z is C_3 0.75

Degree of membership



Example 2

Step 3: Aggregation of rule consequents

Example 2

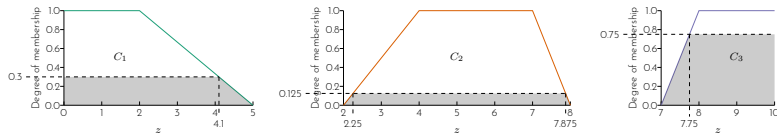
Step 3: Aggregation of rule consequents

The clipped sets of the output variables are combined to form a single fuzzy set.

Example 2

Step 3: Aggregation of rule consequents

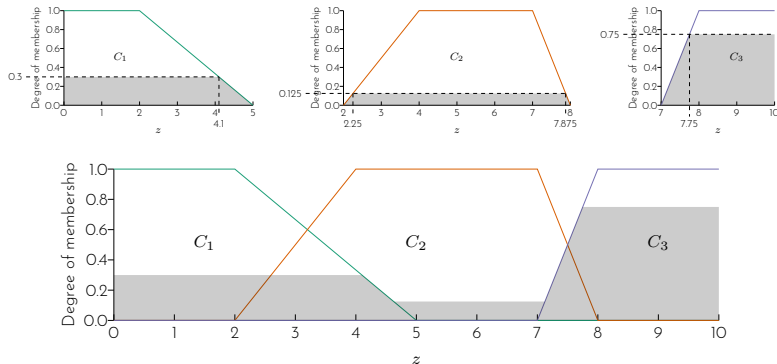
The clipped sets of the output variables are combined to form a single fuzzy set.



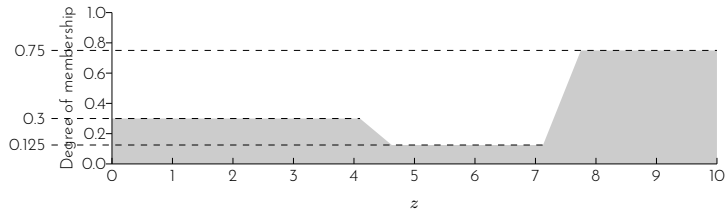
Example 2

Step 3: Aggregation of rule consequents

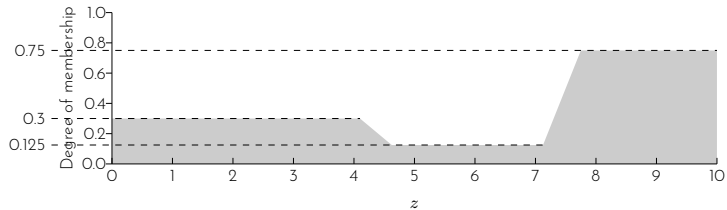
The clipped sets of the output variables are combined to form a single fuzzy set.



Example 2

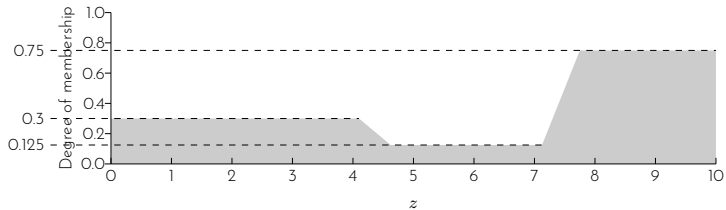


Example 2



Step 4: Defuzzification

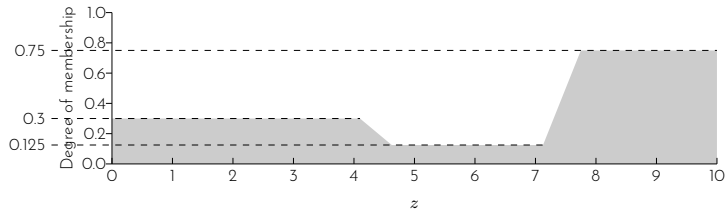
Example 2



Step 4: Defuzzification

Defuzzification for Mamdani-style inference uses the centroid technique:

Example 2

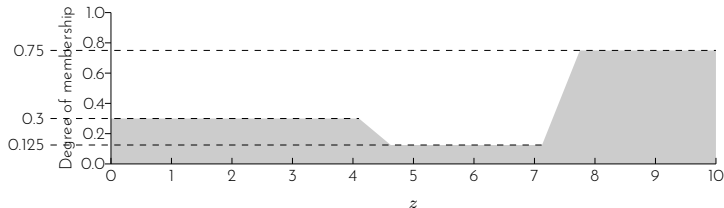


Step 4: Defuzzification

Defuzzification for Mamdani-style inference uses the centroid technique:

$$\text{output} = \frac{\sum_{x=a}^b \mu_A \cdot x}{\sum_{x=a}^b \mu_A}$$

Example 2

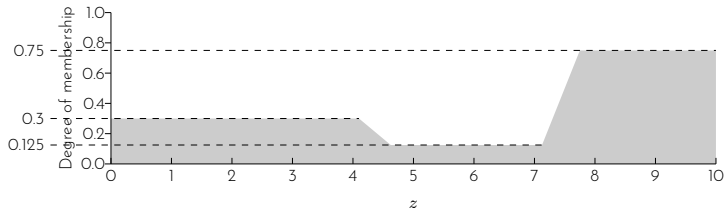


Step 4: Defuzzification

Defuzzification for Mamdani-style inference uses the centroid technique:

$$\text{output} = \frac{\sum_{x=a}^b \mu_A \cdot x}{\sum_{x=a}^b \mu_A} = \frac{(0 + 1 + 2 + 3 + 4)(0.3) + (5 + 6 + 7)(0.125) + (8 + 9 + 10)(0.75)}{5(0.3) + 3(0.125) + 3(0.75)}$$

Example 2

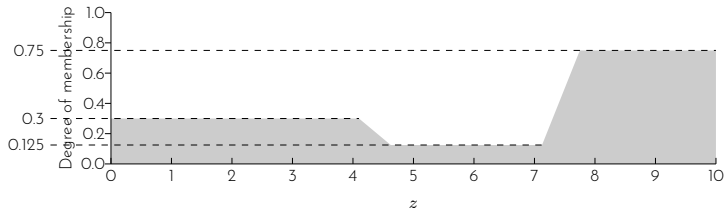


Step 4: Defuzzification

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$$\begin{aligned} \text{output} &= \frac{\sum_{x=a}^b \mu_A \cdot x}{\sum_{x=a}^b \mu_A} = \frac{(0 + 1 + 2 + 3 + 4)(0.3) + (5 + 6 + 7)(0.125) + (8 + 9 + 10)(0.75)}{5(0.3) + 3(0.125) + 3(0.75)} \\ &= 6.182 \end{aligned}$$

Example 2



Step 4: Defuzzification

Defuzzification for Mamdani-style inference uses the centroid technique:

$$\begin{aligned} \text{output} &= \frac{\sum_{x=a}^b \mu_A \cdot x}{\sum_{x=a}^b \mu_A} = \frac{(0 + 1 + 2 + 3 + 4)(0.3) + (5 + 6 + 7)(0.125) + (8 + 9 + 10)(0.75)}{5(0.3) + 3(0.125) + 3(0.75)} \\ &= 6.182 \end{aligned}$$

Therefore, for $x_1 = 3.5$ and $y_1 = 6.5$, $z_1 = 6.182$

Example 3

Consider the fuzzy system with two input variables x and y , and one output variable z . All the variables have the range between 0 and 10.

Variable	Fuzzy sets	Fit vector
x	A_1	$(1/3, 0/5)$
x	A_2	$(0/3, 1/7, 0/8)$
x	A_3	$(0/6, 1/8)$
y	B_1	$(1/3, 0/8)$
y	B_2	$(0/5, 1/8)$

Example 3

The following Sugeno-style rules apply to this system.

Rule 1

IF x is A_3

OR y is B_1

THEN z is 3

Rule 2

IF x is A_2

AND y is B_2

THEN z is 5

Rule 3

IF x is A_1

THEN z is 8

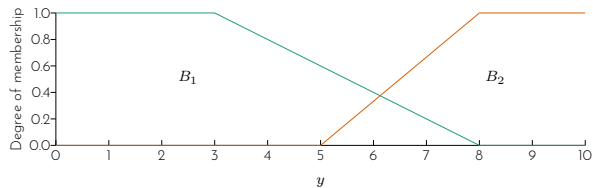
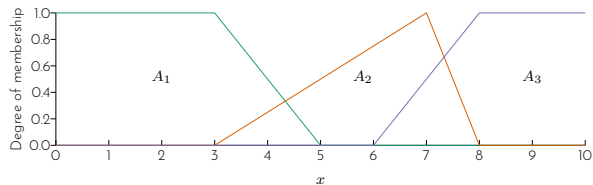
Given that x_1 is 3.5, y_1 is 6.5, what is the value of z_1 ?

Example 3

First, draw the membership function of the fuzzy sets on the graph paper.

Variable	Fuzzy sets	Fit vector
x	A_1	$(1/3, 0/5)$
x	A_2	$(0/3, 1/7, 0/8)$
x	A_3	$(0/6, 1/8)$
y	B_1	$(1/3, 0/8)$
y	B_2	$(0/5, 1/8)$

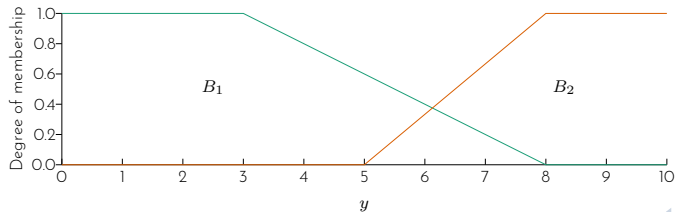
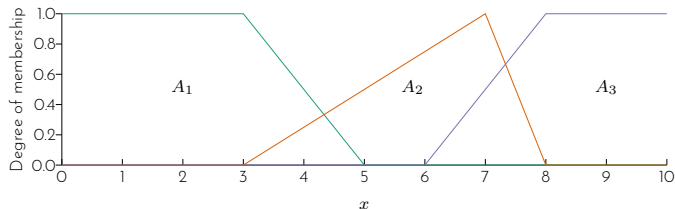
Example 3



Example 3

Step 1: Fuzzification

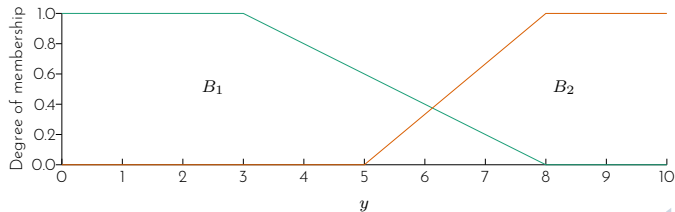
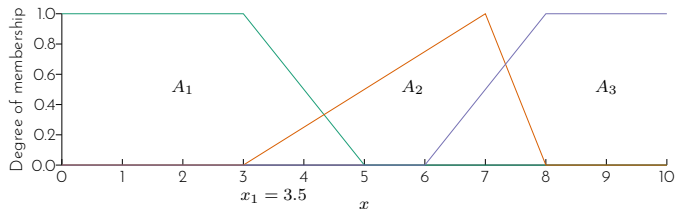
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 3

Step 1: Fuzzification

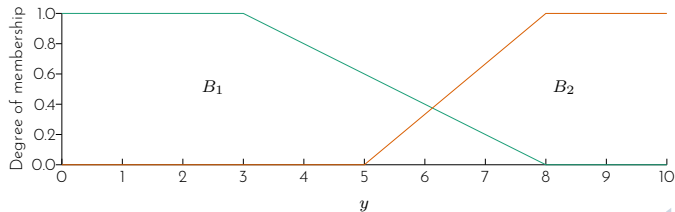
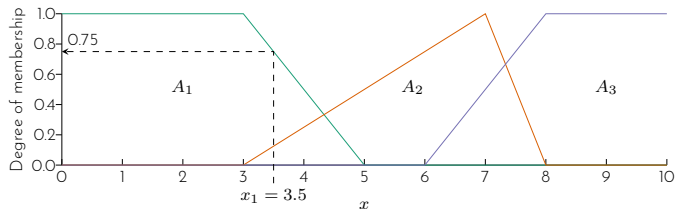
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 3

Step 1: Fuzzification

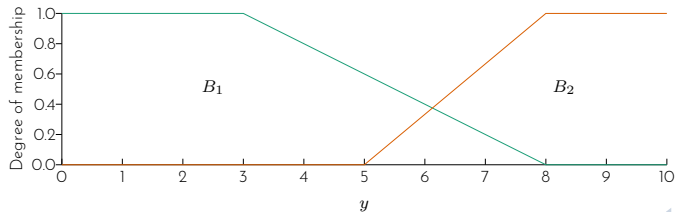
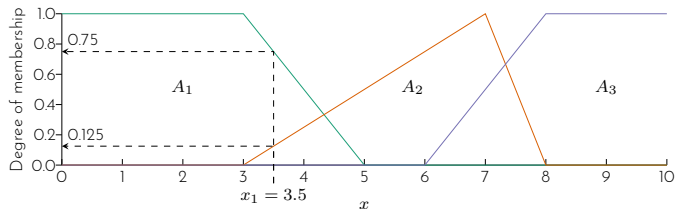
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 3

Step 1: Fuzzification

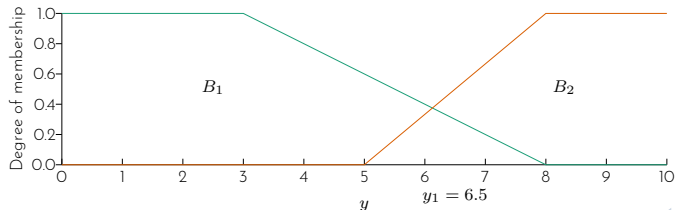
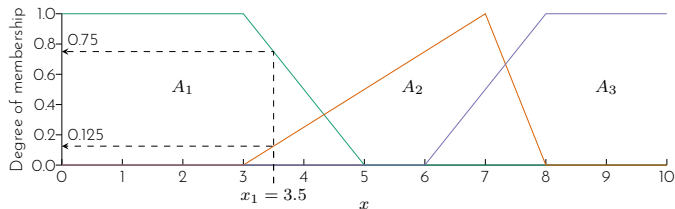
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 3

Step 1: Fuzzification

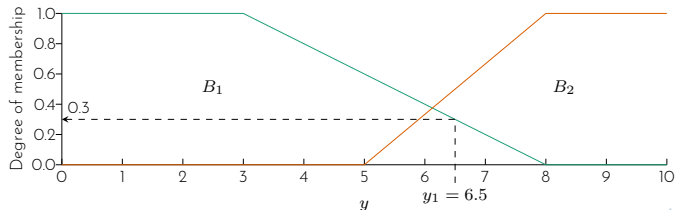
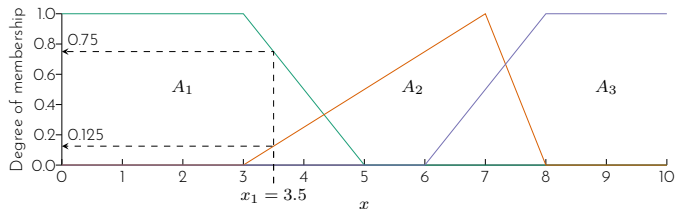
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 3

Step 1: Fuzzification

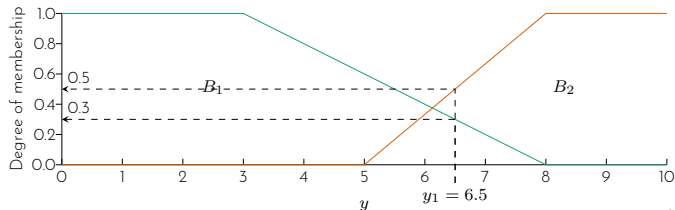
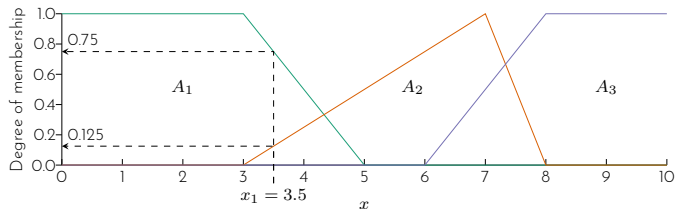
Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 3

Step 1: Fuzzification

Identify the degree of membership for $x_1 = 3.5$ and $y_1 = 6.5$.



Example 3

Step 1: Fuzzification (continued)

Therefore we have

$$\mu_{x=A_1} = 0.75$$

$$\mu_{x=A_2} = 0.125$$

$$\mu_{x=A_3} = 0$$

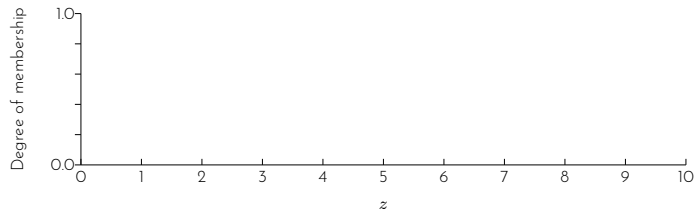
$$\mu_{y=B_1} = 0.3$$

$$\mu_{y=B_2} = 0.5$$

Example 3

Step 2: Rule evaluation

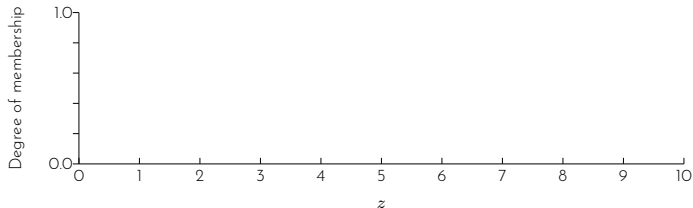
Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is 3	



Example 3

Step 2: Rule evaluation

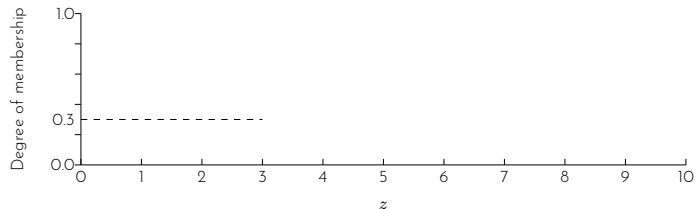
Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is 3	$\max(0.0, 0.3) = 0.3$



Example 3

Step 2: Rule evaluation

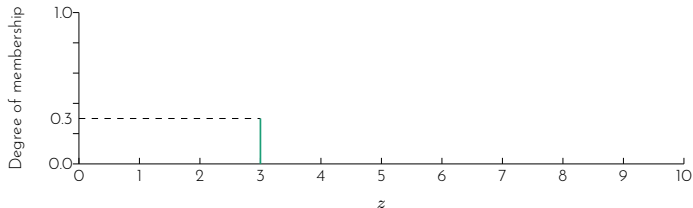
Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is 3	$\max(0.0, 0.3) = 0.3$



Example 3

Step 2: Rule evaluation

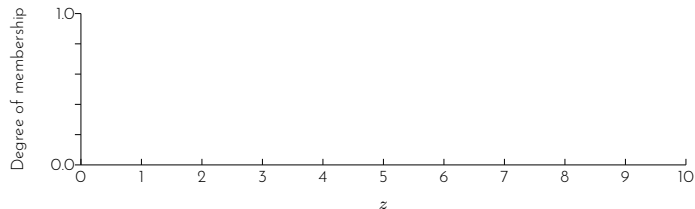
Rule 1		Degree of membership
IF	x is A_3	0.0
OR	y is B_1	0.3
THEN	z is 3	$\max(0.0, 0.3) = 0.3$



Example 3

Step 2: Rule evaluation (continued)

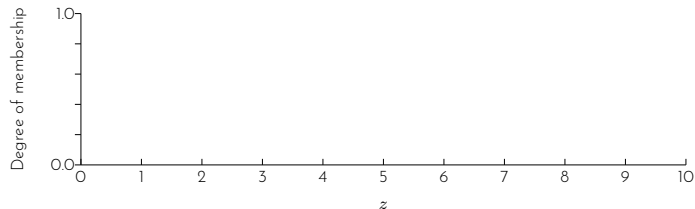
Rule 2		Degree of membership
IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is 5	



Example 3

Step 2: Rule evaluation (continued)

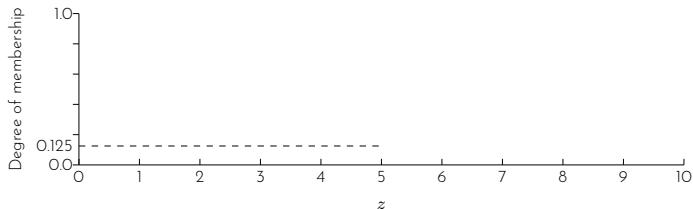
Rule 2		Degree of membership
IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is 5	$\min(0.125, 0.5) = 0.125$



Example 3

Step 2: Rule evaluation (continued)

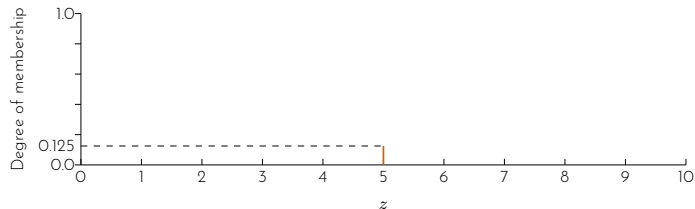
Rule 2		Degree of membership
IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is 5	$\min(0.125, 0.5) = 0.125$



Example 3

Step 2: Rule evaluation (continued)

Rule 2		Degree of membership
IF	x is A_2	0.125
AND	y is B_2	0.5
THEN	z is 5	$\min(0.125, 0.5) = 0.125$



Example 3

Step 2: Rule evaluation (continued)

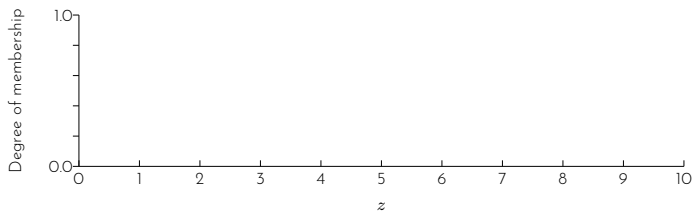
Rule 3

IF x is A_1

THEN z is 8

Degree of membership

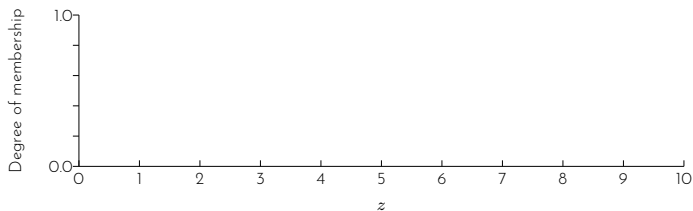
0.75



Example 3

Step 2: Rule evaluation (continued)

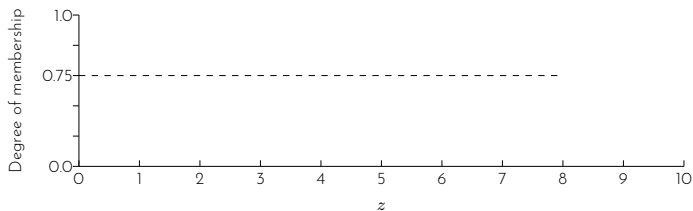
Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is 8	0.75



Example 3

Step 2: Rule evaluation (continued)

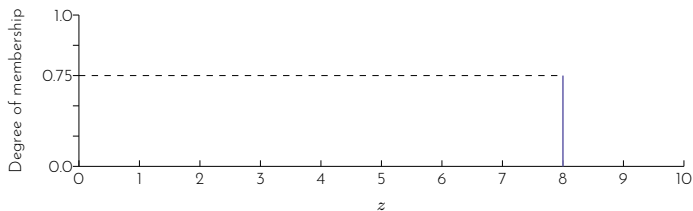
Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is 8	0.75



Example 3

Step 2: Rule evaluation (continued)

Rule 3		Degree of membership
IF	x is A_1	0.75
THEN	z is 8	0.75



Example 3

Step 3: Aggregation of rule consequents

Example 3

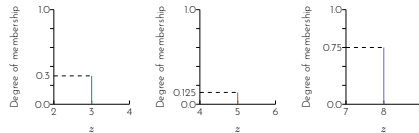
Step 3: Aggregation of rule consequents

The outputs of all fuzzy rules are combined.

Example 3

Step 3: Aggregation of rule consequents

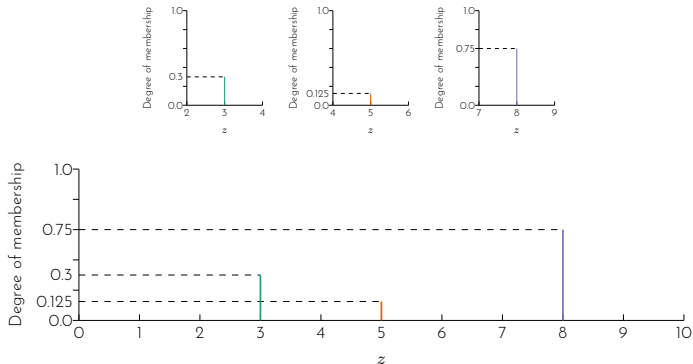
The outputs of all fuzzy rules are combined.



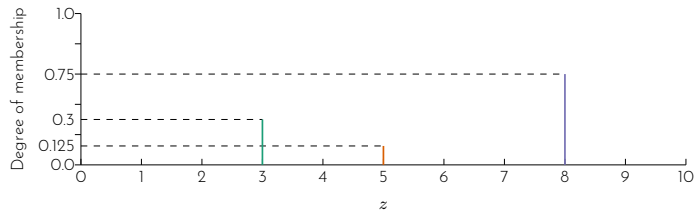
Example 3

Step 3: Aggregation of rule consequents

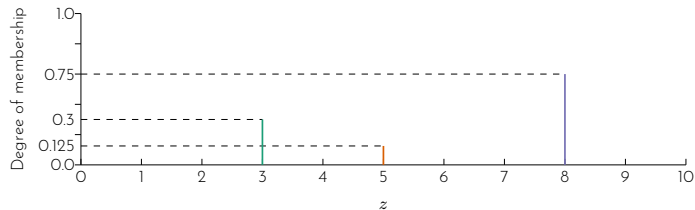
The outputs of all fuzzy rules are combined.



Example 3

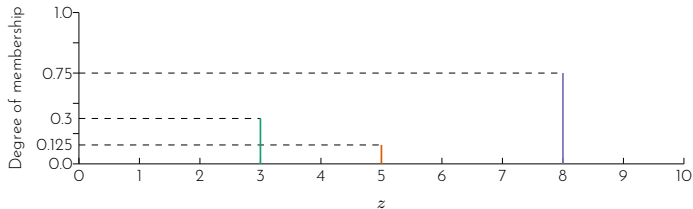


Example 3



Step 4: Defuzzification

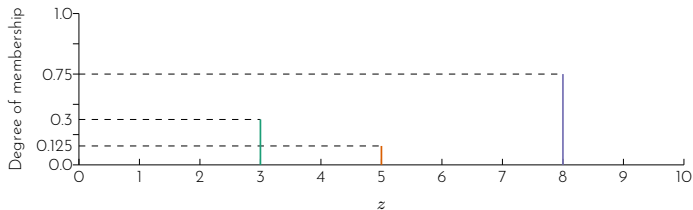
Example 3



Step 4: Defuzzification

Defuzzification for Sugeno-style inference uses the weighted average technique:

Example 3

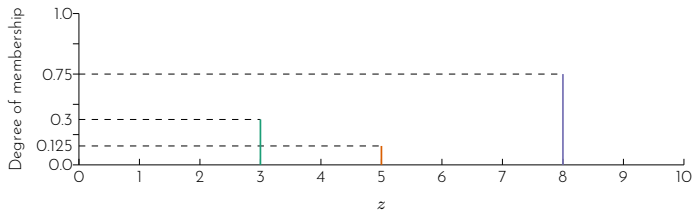


Step 4: Defuzzification

Defuzzification for Sugeno-style inference uses the weighted average technique:

$$\text{output} = \frac{\sum_{x=a}^b \alpha_x \cdot x}{\sum_{x=a}^b \alpha_x}$$

Example 3

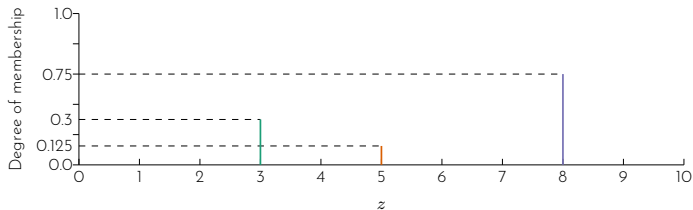


Step 4: Defuzzification

Defuzzification for Sugeno-style inference uses the weighted average technique:

$$\text{output} = \frac{\sum_{x=a}^b \alpha_x \cdot x}{\sum_{x=a}^b \alpha_x} = \frac{(3)(0.3) + (5)(0.125) + (8)(0.75)}{0.3 + 0.125 + 0.75}$$

Example 3

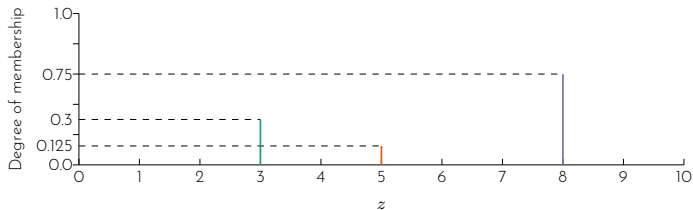


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Defuzzification for Sugeno-style inference uses the weighted average technique:

$$\begin{aligned}\text{output} &= \frac{\sum_{x=a}^b \alpha_x \cdot x}{\sum_{x=a}^b \alpha_x} = \frac{(3)(0.3) + (5)(0.125) + (8)(0.75)}{0.3 + 0.125 + 0.75} \\ &= 6.404\end{aligned}$$

Example 3



Step 4: Defuzzification

Defuzzification for Sugeno-style inference uses the weighted average technique:

$$\begin{aligned}\text{output} &= \frac{\sum_{x=a}^b \alpha_x \cdot x}{\sum_{x=a}^b \alpha_x} = \frac{(3)(0.3) + (5)(0.125) + (8)(0.75)}{0.3 + 0.125 + 0.75} \\ &= 6.404\end{aligned}$$

Therefore, for $x_1 = 3.5$ and $y_1 = 6.5$, $z_1 = 6.404$

What are the variables that can be represented as fuzzy variables?

Building a fuzzy expert system

The typical process in developing a fuzzy expert system incorporates the following steps:

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1. Specify the problem and define linguistic variables.

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4. Encode the fuzzy sets, fuzzy rules, and procedures to perform fuzzy inference into the expert system.

The typical process in developing a fuzzy expert system incorporates the following steps:

1. Specify the problem and define linguistic variables.
2. Determine fuzzy sets.
3. Elicit and construct fuzzy rules.
4. Encode the fuzzy sets, fuzzy rules, and procedures to perform fuzzy inference into the expert system.
5. Evaluate and tune the system.

Case study 4

Build a fuzzy expert system to control the speed of a train.

Case study 4

Build a fuzzy expert system to control the speed of a train.

Step 1: Specify the problem and define linguistic variables.

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Nurturing the Seeds of Wisdom

Objective:

Design system to smoothly slow and stop train from any speed and distance from station.

Linguistic variables:

Case study 4

Build a fuzzy expert system to control the speed of a train.

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Objective:

Design system to smoothly slow and stop train from any speed and distance from station.

Linguistic variables:

Input

Case study 4

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Step 1: Specify the problem and define linguistic variables.

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Design system to smoothly slow and stop train from any speed and distance from station.

Linguistic variables:

Input – speed, distance

Created with L^AT_EX 53

Build a fuzzy expert system to control the speed of a train.

Step 1: Specify the problem and define linguistic variables.

Objective:

Design system to smoothly slow and stop train from any speed and distance from station.

Linguistic variables:

Input – speed, distance

Output – brake, throttle

Case study 4

Step 1 (continued) Define the range of the linguistic variables.

Case study 4

Step 1 (continued) Define the range of the linguistic variables.

Speed	
Linguistic value	Numerical range
Fast	26.5 - 70
Medium fast	6.5 - 46.5
Slow	2.5 - 10.5
Very slow	1 - 4
Stopped	0 - 2

Case study 4

Step 1 (continued) Define the range of the linguistic variables.

Speed	
Linguistic value	Numerical range
Fast	26.5 - 70
Medium fast	6.5 - 46.5
Slow	2.5 - 10.5
Very slow	1 - 4
Stopped	0 - 2

Distance	
Linguistic value	Numerical range
Far	1500 - ∞
Medium far	100 - 3000
Near	3 - 200
Very near	1 - 5
At	0 - 2

Case study 4

Step 1 (continued) Define the range of the linguistic variables.

Speed	
Linguistic value	Numerical range
Fast	26.5 - 70
Medium fast	6.5 - 46.5
Slow	2.5 - 10.5
Very slow	1 - 4
Stopped	0 - 2

Distance	
Linguistic value	Numerical range
Far	1500 - ∞
Medium far	100 - 3000
Near	3 - 200
Very near	1 - 5
At	0 - 2

Throttle	
Linguistic value	Numerical range
Full	60% - 100%
Medium	20% - 80%
Slight	3% - 30%
Very slight	1% - 5%
No	0% - 2%

Case study 4

Step 1 (continued) Define the range of the linguistic variables.

Speed	
Linguistic value	Numerical range
Fast	26.5 - 70
Medium fast	6.5 - 46.5
Slow	2.5 - 10.5
Very slow	1 - 4
Stopped	0 - 2

Distance	
Linguistic value	Numerical range
Far	1500 - ∞
Medium far	100 - 3000
Near	3 - 200
Very near	1 - 5
At	0 - 2

Throttle	
Linguistic value	Numerical range
Full	60% - 100%
Medium	20% - 80%
Slight	3% - 30%
Very slight	1% - 5%
No	0% - 2%

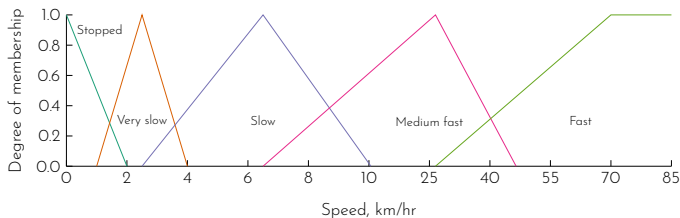
Brake	
Linguistic value	Numerical range
Full	98% - 100%
Medium	95% - 99%
Slight	70% - 97%
Very slight	20% - 80%
No	0% - 40%

Case study 4

Step 2: Determine fuzzy sets.

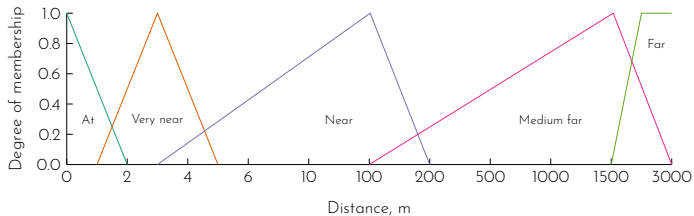
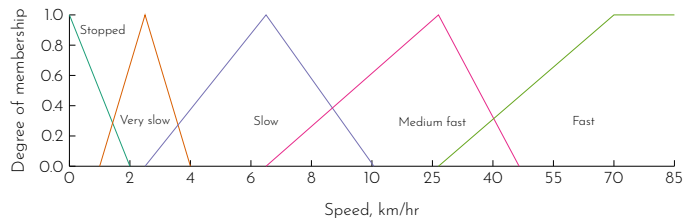
Case study 4

Step 2: Determine fuzzy sets.



Case study 4

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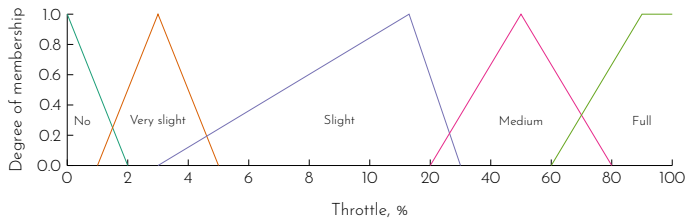


Case study 4

Step 2 (continued)

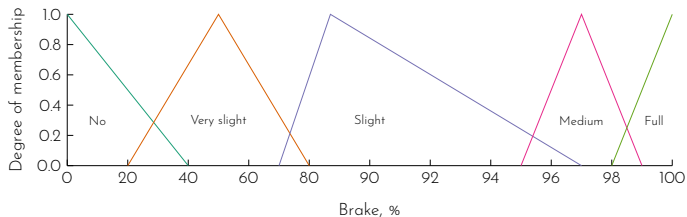
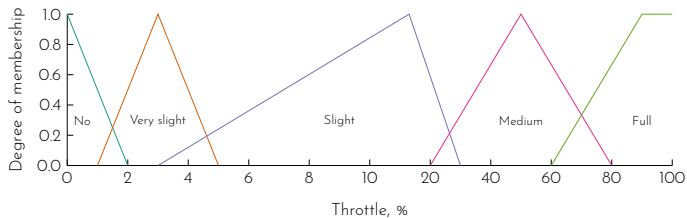
Case study 4

Step 2 (continued)



Case study 4

Step 2 (continued)



Case study 4

Step 3: Elicit and construct fuzzy rules.

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Fuzzy association memory (FAM) representation

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Fuzzy association memory (FAM) representation

- Represent fuzzy rules in matrix form.

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- ▶ Represent fuzzy rules in matrix form.
- ▶ Each dimension of the matrix represents an input variable.

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- ▶ Each dimension of the matrix represents an input variable.
- ▶ The number of row/column along each dimension equals to the number of subsets in the corresponding input variable.

Step 3: Elicit and construct fuzzy rules.

Fuzzy association memory (FAM) representation

- ▶ Represent fuzzy rules in matrix form.
- ▶ Each dimension of the matrix represents an input variable.
- ▶ The number of row/column along each dimension equals to the number of subsets in the corresponding input variable.
- ▶ The elements in each cell is the rule consequent.

Case study 4

Step 3 (continued)

Case study 4

Step 3 (continued)

		Distance				
		At	Very near	Near	Medium far	Far
Speed	Stopped					
	Very slow					
	Slow					
	Medium fast					
	Fast					

Case study 4

Step 3 (continued)

		Distance				
		At	Very near	Near	Medium far	Far
Speed	Stopped					
	Very slow					
	Slow					
	Medium fast					
	Fast					

Each cell provides the rule consequent for the intersection (AND) of the corresponding input values.

Case study 4

Step 3 (continued)

		Distance				
		At	Very near	Near	Medium far	Far
Speed	Stopped					
	Very slow					
	Slow					
	Medium fast					
	Fast					

Each cell provides the rule consequent for the intersection (AND) of the corresponding input values.

For example, the top left cell represents the rule consequent of 'speed' is 'stopped' AND 'distance' is 'at'.

Case study 4

Step 3 (continued)

		Distance				
		At	Very near	Near	Medium far	Far
Speed	Stopped	Full brake No throttle	Full brake VS throttle			
	Very slow	Full brake No throttle	Medium brake VS throttle	Slight brake VS throttle		
	Slow	Full brake No throttle	Medium brake VS throttle	VS brake Slight throttle		
	Medium fast				VS brake Med. throttle	No brake Full throttle
	Fast				VS brake Med. throttle	No brake Full throttle

Step 4: Encode the fuzzy sets, fuzzy rules, and procedures to perform fuzzy inference into the expert system.

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- ▶ This step concerns of actually building the expert system using a library or toolbox.

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- ▶ This step concerns of actually building the expert system using a library or toolbox.
- ▶ It also includes making decision on how the rules are evaluated. In most cases, we will be using the centroid method.

Case study 4

Step 4 (continued)

To illustrate how the rules are evaluated, we consider the inputs of

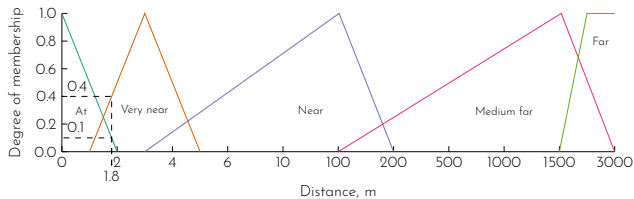
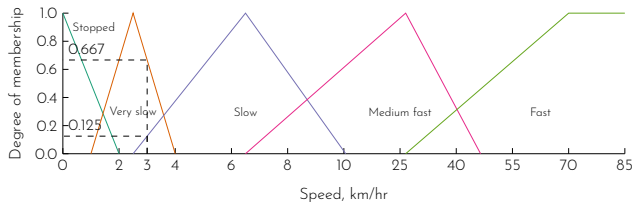
speed = 3 km/hr

distance = 1.8 m

Case study 4

Step 4 (continued)

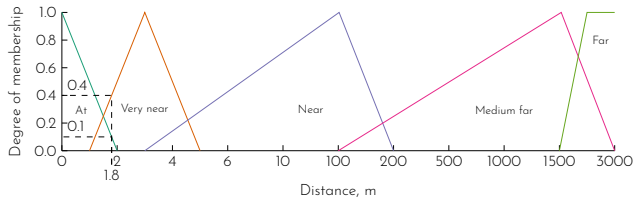
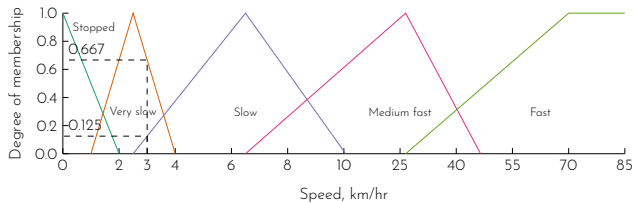
With speed = 3 km/hr and distance = 1.8 m,



Case study 4

Step 4 (continued)

With speed = 3 km/hr and distance = 1.8 m,



From the fuzzy set diagrams, we have

$$\mu_{\text{speed}=\text{stopped}} = 0$$

$$\mu_{\text{speed}=\text{very slow}} = 0.667$$

$$\mu_{\text{speed}=\text{slow}} = 0.125$$

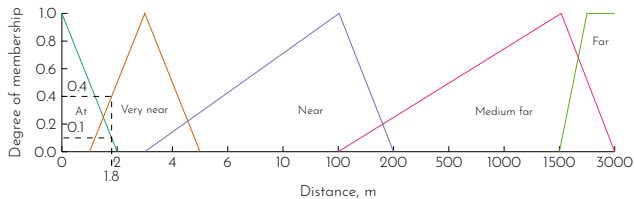
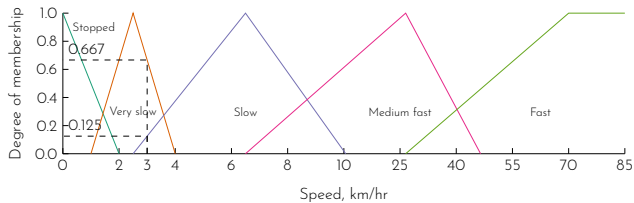
$$\mu_{\text{speed}=\text{medium fast}} = 0$$

$$\mu_{\text{speed}=\text{fast}} = 0$$

Case study 4

Step 4 (continued)

With speed = 3 km/hr and distance = 1.8 m,



From the fuzzy set diagrams, we have

$$\mu_{\text{speed}=\text{stopped}} = 0$$

$$\mu_{\text{speed}=\text{very slow}} = 0.667$$

$$\mu_{\text{speed}=\text{slow}} = 0.125$$

$$\mu_{\text{speed}=\text{medium fast}} = 0$$

$$\mu_{\text{speed}=\text{fast}} = 0$$

$$\mu_{\text{distance}=\text{at}} = 0.1$$

$$\mu_{\text{distance}=\text{very near}} = 0.4$$

$$\mu_{\text{distance}=\text{near}} = 0$$

$$\mu_{\text{distance}=\text{medium far}} = 0$$

$$\mu_{\text{distance}=\text{far}} = 0$$

Case study 4

Step 4 (continued)

Case study 4

Step 4 (continued)

Since we only have non-zero degree of membership for

speed = very slow

speed = slow

$$\text{distance} = at$$

distance = very near

Case study 4

Step 4 (continued)

Since we only have non-zero degree of membership for

speed = very slow

speed = slow

distance = at

distance = very near

Only the shaded rules are fired.

		Distance				
		At	Very near	Near	Medium far	Far
Speed	Stopped	Full brake No throttle	Full brake VS throttle			
	Very slow	Full brake No throttle	Medium brake VS throttle	Slow brake VS throttle		
		Full brake No throttle	Medium brake VS throttle	VS brake Slow throttle		
	Slow					
	Medium fast				VS brake Med. throttle	No brake Full throttle
	Fast				VS brake Med. throttle	No brake Full throttle

Case study 4

Step 4 (continued)

Case study 4

Step 4 (continued)

The four rules that fired are

Step 4 (continued)

The four rules that fired are

Rule 1

IF	speed = very slow	0.667
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

Step 4 (continued)

The four rules that fired are

Rule 1

IF	speed = very slow	0.667
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

Rule 2

IF	speed = very slow	0.667
AND	distance = very near	0.4
THEN	brake = medium	0.4
	throttle = very slight	0.4

Step 4 (continued)

The four rules that fired are

Rule 1

IF	speed = very slow	0.667
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

Rule 2

IF	speed = very slow	0.667
AND	distance = very near	0.4
THEN	brake = medium	0.4
	throttle = very slight	0.4

Rule 3

IF	speed = slow	0.125
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

Step 4 (continued)

The four rules that fired are

Rule 1

IF	speed = very slow	0.667
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

Rule 2

IF	speed = very slow	0.667
AND	distance = very near	0.4
THEN	brake = medium	0.4
	throttle = very slight	0.4

Rule 3

IF	speed = slow	0.125
AND	distance = at	0.1
THEN	brake = full	0.1
	throttle = no	0.1

Rule 4

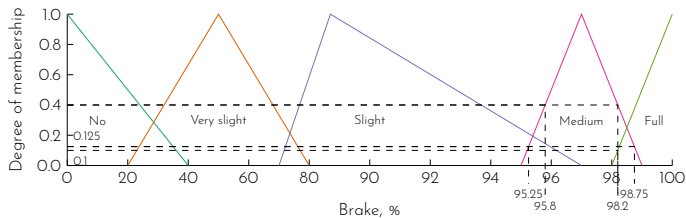
IF	speed = slow	0.125
AND	distance = very near	0.4
THEN	brake = medium	0.125
	throttle = very slight	0.125

Case study 4

Step 4 (continued)

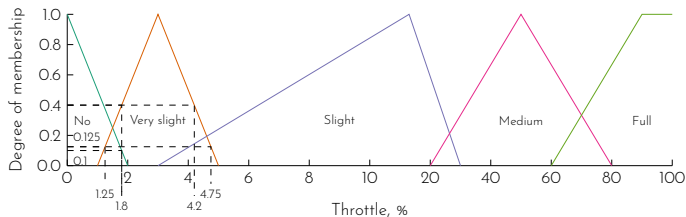
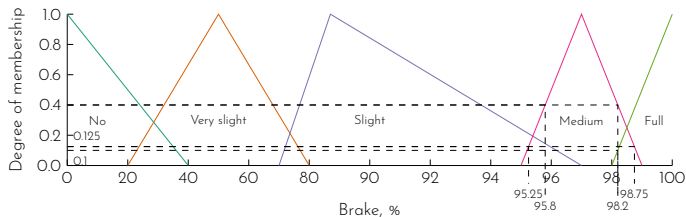
Case study 4

Step 4 (continued)



Case study 4

Step 4 (continued)



Case study 4

Step 4 (continued)

Case study 4

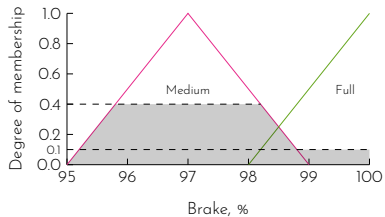
Step 4 (continued)

Aggregation of outputs

Case study 4

Step 4 (continued)

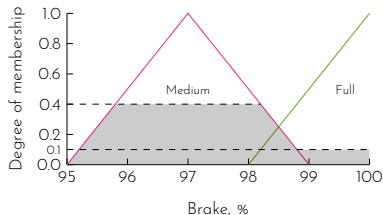
Aggregation of outputs



Case study 4

Step 4 (continued)

Aggregation of outputs



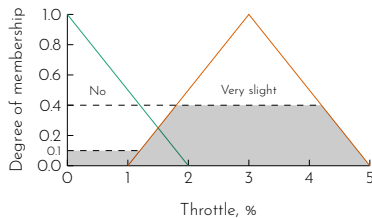
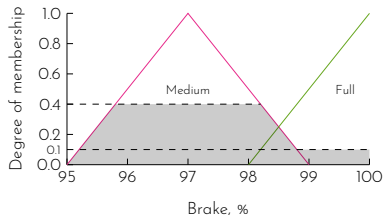
brake

$$\begin{aligned} & (95)(0) \\ & + (96 + 97 + 98)(0.4) \\ & + (99 + 100)(0.1) \\ = & \frac{0 + 3(0.4) + 2(0.1)}{1} \\ = & 97.36\% \end{aligned}$$

Case study 4

Step 4 (continued)

Aggregation of outputs



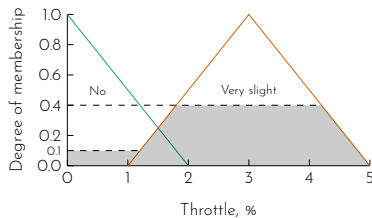
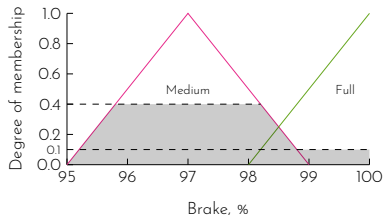
brake

$$\begin{aligned} & (95)(0) \\ & + (96 + 97 + 98)(0.4) \\ & + (99 + 100)(0.1) \\ = & \frac{0 + 3(0.4) + 2(0.1)}{0 + 3(0.4) + 2(0.1)} \\ = & 97.36\% \end{aligned}$$

Case study 4

Step 4 (continued)

Aggregation of outputs



brake

$$\begin{aligned} & (95)(0) \\ & + (96 + 97 + 98)(0.4) \\ & + (99 + 100)(0.1) \\ = & \frac{0 + 3(0.4) + 2(0.1)}{0 + 3(0.4) + 2(0.1)} \\ = & 97.36\% \end{aligned}$$

throttle

$$\begin{aligned} & (0 + 1)(0.1) \\ & + (2 + 3 + 4)(0.4) \\ & + (5)(0) \\ = & \frac{2(0.1) + 3(0.4) + 0}{2(0.1) + 3(0.4) + 0} \\ = & 2.64\% \end{aligned}$$

Case study 4

Step 5: Evaluate and tune the system

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- ▶ This step is to identify if the designed fuzzy system meets the requirements of the specified problem.

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- ▶ Evaluation can be carried out by identifying the outputs of several test situations where different values of inputs are provided.

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- ▶ Evaluation can be carried out by identifying the outputs of several test situations where different values of inputs are provided.
- ▶ Another method of evaluation is to generate surface plots of the rules. This can be achieved using appropriate tools.

Step 5: Evaluate and tune the system

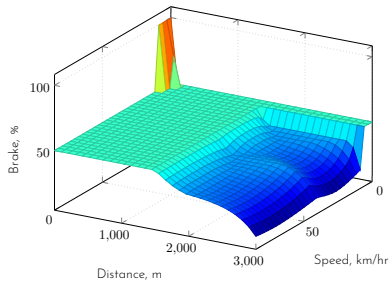
- ▶ This step is to identify if the designed fuzzy system meets the requirements of the specified problem.
- ▶ Evaluation can be carried out by identifying the outputs of several test situations where different values of inputs are provided.
- ▶ Another method of evaluation is to generate surface plots of the rules. This can be achieved using appropriate tools.
- ▶ What parameters of the system are we tuning?

Case study 4

Step 5 (continued)

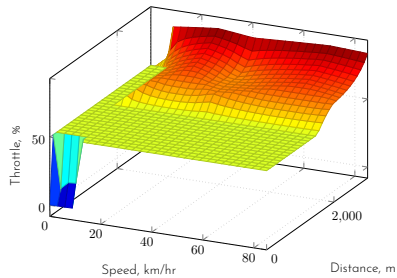
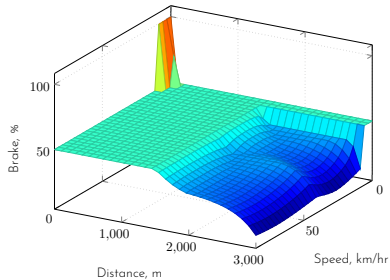
Case study 4

Step 5 (continued)



Case study 4

Step 5 (continued)



Case study 5

Build a fuzzy expert system to identify the percentage of tips a customer will give based on the service and the food the customer received.

Case study 5

Step 1: Specify the problem and define linguistic variables

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Objective:

Design a system to identify the percentage of tips a customer based on the service and food.

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Objective:

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Linguistic variables:

Step 1: Specify the problem and define linguistic variables

Objective:

Design a system to identify the percentage of tips a customer based on the service and food.

Linguistic variables:

Input

Step 1: Specify the problem and define linguistic variables

Objective:

Design a system to identify the percentage of tips a customer based on the service and food.

Linguistic variables:

Input – service, food

Step 1: Specify the problem and define linguistic variables

Objective:

Design a system to identify the percentage of tips a customer based on the service and food.

Linguistic variables:

Input – service, food

Output

Step 1: Specify the problem and define linguistic variables

Objective:

Design a system to identify the percentage of tips a customer based on the service and food.

Linguistic variables:

Input - service, food

Output - tips

Case study 5

Step 1 (continued)

Case study 5

Step 1 (continued)

Define the range of the linguistic variables.

Case study 5

Step 1 (continued)

Define the range of the linguistic variables.

Service	
Linguistic value	Numerical range
Poor	0 - 5
Average	0 - 10
Good	5 - 10

Case study 5

Step 1 (continued)

Define the range of the linguistic variables.

Service	
Linguistic value	Numerical range
Poor	0 - 5
Average	0 - 10
Good	5 - 10

Food	
Linguistic value	Numerical range
Poor	0 - 5
Average	0 - 10
Good	5 - 10

Case study 5

Step 1 (continued)

Define the range of the linguistic variables.

Service	
Linguistic value	Numerical range
Poor	0 - 5
Average	0 - 10
Good	5 - 10

Food	
Linguistic value	Numerical range
Poor	0 - 5
Average	0 - 10
Good	5 - 10

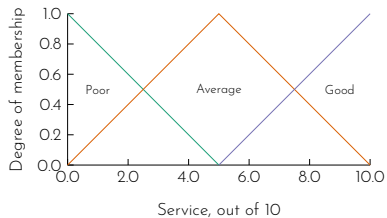
Tips	
Linguistic value	Numerical range
Low	0% - 15%
Medium	0% - 30%
High	15% - 30%

Case study 5

Step 2: Determine fuzzy sets.

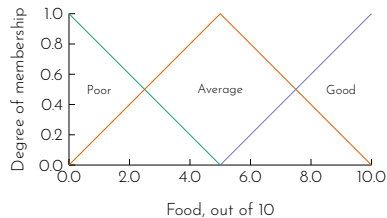
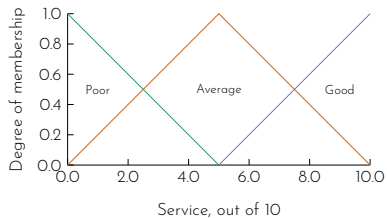
Case study 5

Step 2: Determine fuzzy sets.



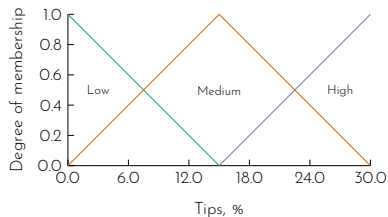
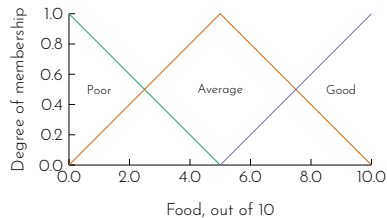
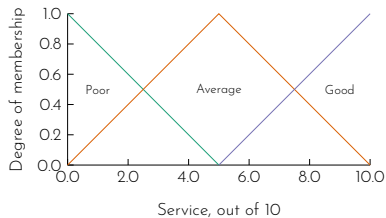
Case study 5

Step 2: Determine fuzzy sets.



Case study 5

Step 2: Determine fuzzy sets.



Case study 5

Step 3: Elicit and construct fuzzy rules.

Case study 5

Step 3: Elicit and construct fuzzy rules.

		Food		
		Poor	Average	Good
Service	Poor	low tips	low tips	medium tips
	Average	low tips	medium tips	high tips
	Good	medium tips	high tips	high tips

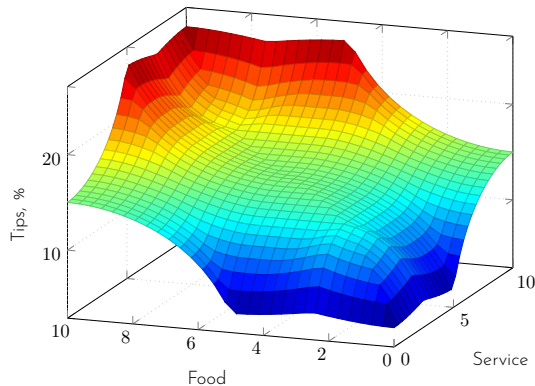
Step 4: Encode the fuzzy sets, fuzzy rules, and procedures to perform fuzzy inference into the expert system.

We can use Matlab Fuzzy Logic Toolbox to encode our fuzzy system.

Case study 5

Step 5: Evaluate and tune the system

The outputs of the fuzzy system are plotted against the range of the inputs (0-10).



Case study 5

Step 5 (continued)

Case study 5

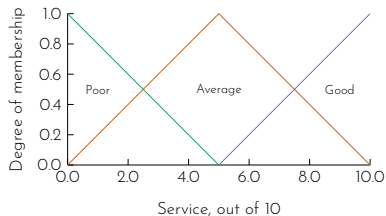
Step 5 (continued)

If I'm not happy with the result, I might modify the degree of membership in the score of service such that

Case study 5

Step 5 (continued)

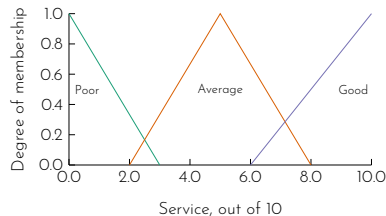
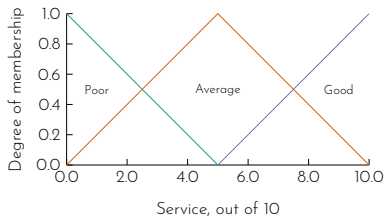
If I'm not happy with the result, I might modify the degree of membership in the score of service such that



Case study 5

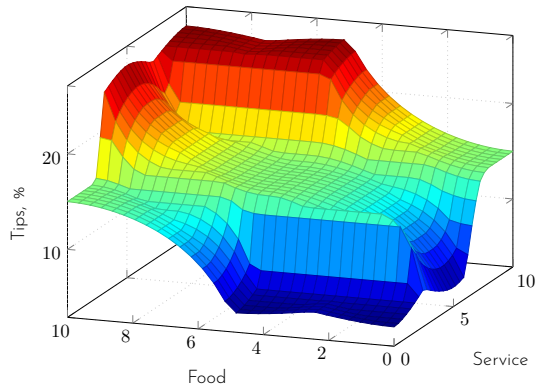
Step 5 (continued)

If I'm not happy with the result, I might modify the degree of membership in the score of service such that



Case study 5

Step 5 (continued)



Case study 6

Consider a problem of operating a service centre of spare parts.

A service centre keeps spare parts and repairs failed ones. A customer brings a failed item and receives a spare of the same type. Failed parts are repaired, placed on the shelf, and thus become spares. If the required spare is available on the shelf, the customer takes it and leaves the service centre. However, if there is no spare on the shelf, the customer has to wait until the needed item becomes available. The objective here is to advise a manager of the service centre on certain decision policies to keep the customers satisfied.

Case study 6

Step 1: Specify the problem and define linguistic variables

Case study 6

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

Linguistic variables:

- A set of small navigation icons typically found in Beamer presentations, including symbols for back, forward, search, and other slide controls.

Case study 6

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

- ▶ mean delay - the average waiting time of a customer.
- ▶ number of servers - the number of people in customer service.

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

- ▶ mean delay - the average waiting time of a customer.
- ▶ number of servers - the number of people in customer service.
- ▶ utilisation factor - the ratio of the rate of failures to the rate of repairs.

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

- ▶ mean delay - the average waiting time of a customer.
- ▶ number of servers - the number of people in customer service.
- ▶ utilisation factor - the ratio of the rate of failures to the rate of repairs.
- ▶ number of spare parts - the number of spare parts the centre has.

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

- ▶ mean delay - the average waiting time of a customer.
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Input :

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

- ▶ mean delay - the average waiting time of a customer.
- ▶ number of servers - the number of people in customer service.
- ▶ utilisation factor - the ratio of the rate of failures to the rate of repairs.
- ▶ number of spare parts - the number of spare parts the centre has.

Input : mean delay, utilisation factor, number of servers

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

- ▶ mean delay - the average waiting time of a customer.
- ▶ number of servers - the number of people in customer service.
- ▶ utilisation factor - the ratio of the rate of failures to the rate of repairs.
- ▶ number of spare parts - the number of spare parts the centre has.

Input : mean delay, utilisation factor, number of servers

Output :

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

- ▶ mean delay - the average waiting time of a customer.
- ▶ number of servers - the number of people in customer service.
- ▶ utilisation factor - the ratio of the rate of failures to the rate of repairs.
- ▶ number of spare parts - the number of spare parts the centre has.

Input : mean delay, utilisation factor, number of servers

Output : number of spare parts

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

- ▶ mean delay - the average waiting time of a customer.
- ▶ number of servers - the number of people in customer service.
- ▶ utilisation factor - the ratio of the rate of failures to the rate of repairs.
- ▶ number of spare parts - the number of spare parts the centre has.

Input : mean delay, utilisation factor, number of servers

Output : number of spare parts

Objective :

Step 1: Specify the problem and define linguistic variables

Linguistic variables:

- ▶ mean delay - the average waiting time of a customer.
- ▶ number of servers - the number of people in customer service.
- ▶ utilisation factor - the ratio of the rate of failures to the rate of repairs.
- ▶ number of spare parts - the number of spare parts the centre has.

Input : mean delay, utilisation factor, number of servers

Output : number of spare parts

Objective : Advise the manager of the service centre of the number of spares required to maintain the mean delay in customer service within an acceptable range.

Case study 6

Step 1 (continued)

Case study 6

Step 1 (continued)

Define the range of the linguistic variables.

Case study 6

Step 1 (continued)

Define the range of the linguistic variables.

Mean delay (normalised), m	
Linguistic value	Numerical range
very short	0 - 0.3
short	0.1 - 0.5
medium	0.4 - 1

Case study 6

Step 1 (continued)

Define the range of the linguistic variables.

Mean delay (normalised), m	
Linguistic value	Numerical range
very short	0 - 0.3
short	0.1 - 0.5
medium	0.4 - 1

Number of servers (normalised), s	
Linguistic value	Numerical range
small	0 - 0.35
medium	0.3 - 0.7
large	0.6 - 1

Case study 6

Step 1 (continued)

Define the range of the linguistic variables.

Mean delay (normalised), m	
Linguistic value	Numerical range
very short	0 - 0.3
short	0.1 - 0.5
medium	0.4 - 1

Number of servers (normalised), s	
Linguistic value	Numerical range
small	0 - 0.35
medium	0.3 - 0.7
large	0.6 - 1

Utilisation factor, ρ	
Linguistic value	Numerical range
low	0 - 0.6
medium	0.4 - 0.8
high	0.6 - 1

Case study 6

Step 1 (continued)

Define the range of the linguistic variables.

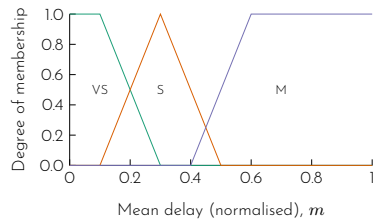
Number of spares (normalised), n	
Linguistic value	Numerical range
very small	0 - 0.30
small	0 - 0.40
rather small	0.25 - 0.45
medium	0.30 - 0.70
rather large	0.55 - 0.75
large	0.60 - 1
very large	0.70 - 1

Case study 6

Step 2: Determine fuzzy sets.

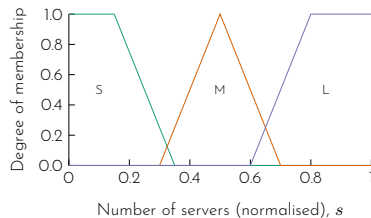
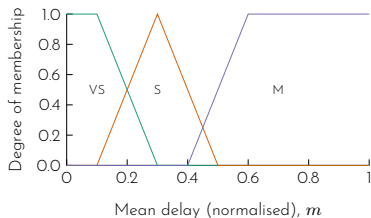
Case study 6

Step 2: Determine fuzzy sets.



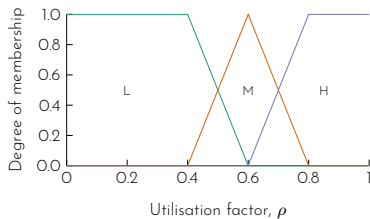
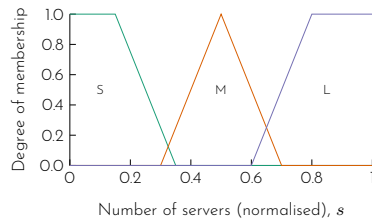
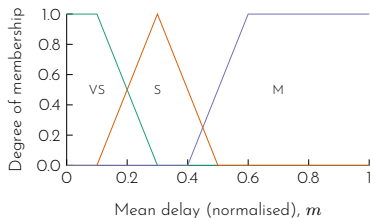
Case study 6

Step 2: Determine fuzzy sets.



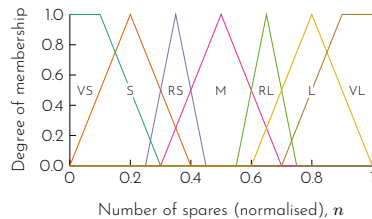
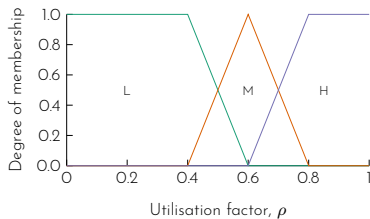
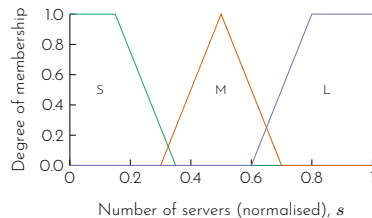
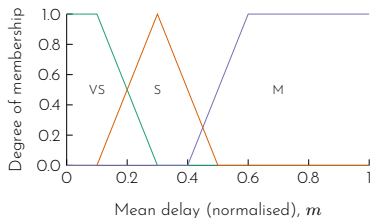
Case study 6

Step 2: Determine fuzzy sets.



Case study 6

Step 2: Determine fuzzy sets.



Case study 6

Step 3: Elicit and construct fuzzy rules.

Case study 6

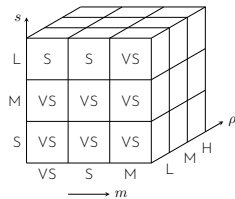
Step 3: Elicit and construct fuzzy rules.

As there are three inputs and one output, we can construct a fuzzy associative memory. The representation takes the shape of $3 \times 3 \times 3$ cube.

Case study 6

Step 3: Elicit and construct fuzzy rules.

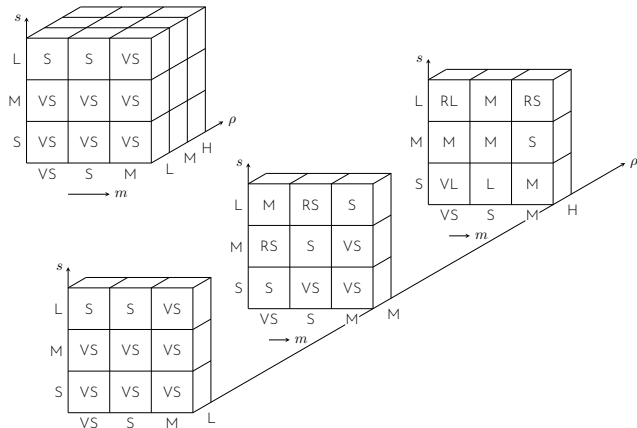
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Case study 6

Step 3: Elicit and construct fuzzy rules.

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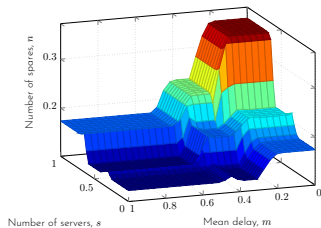
Step 4: Encode the fuzzy sets, fuzzy rules, and procedures to perform fuzzy inference into the expert system.

Case study 6

Step 5: Evaluate and tune the system

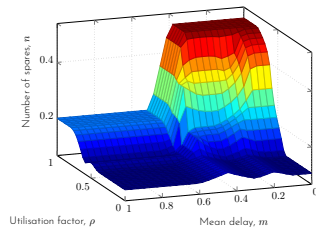
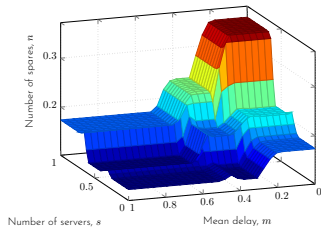
Case study 6

Step 5: Evaluate and tune the system



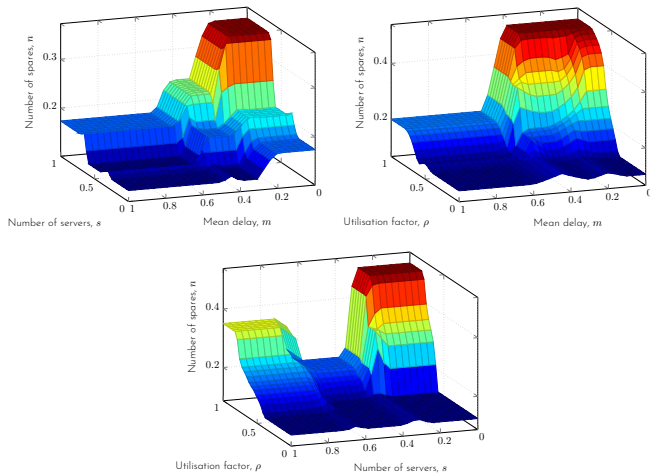
Case study 6

Step 5: Evaluate and tune the system



Case study 6

Step 5: Evaluate and tune the system



Case study 6

Step 5 (continued)

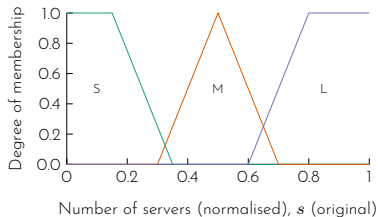
Step 5 (continued)

At this point, the expert might not be satisfied with the system performance. To improve it, he or she may suggest additional sets - 'Rather small' and 'Rather large' for 'number of servers, s '.

Case study 6

Step 5 (continued)

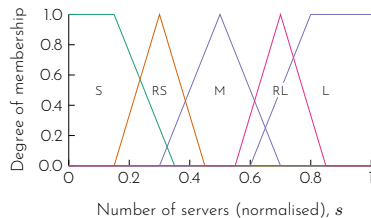
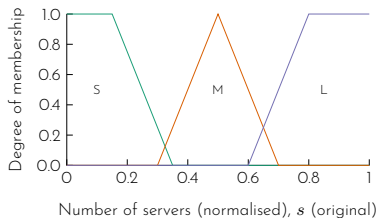
At this point, the expert might not be satisfied with the system performance. To improve it, he or she may suggest additional sets - 'Rather small' and 'Rather large' for 'number of servers, s '.



Case study 6

Step 5 (continued)

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Case study 6

Step 5 (continued)

Case study 6

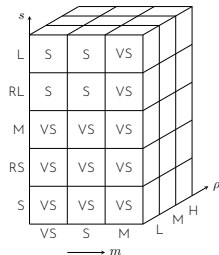
Step 5 (continued)

The rules are updated to include the two new fuzzy sets.

Case study 6

Step 5 (continued)

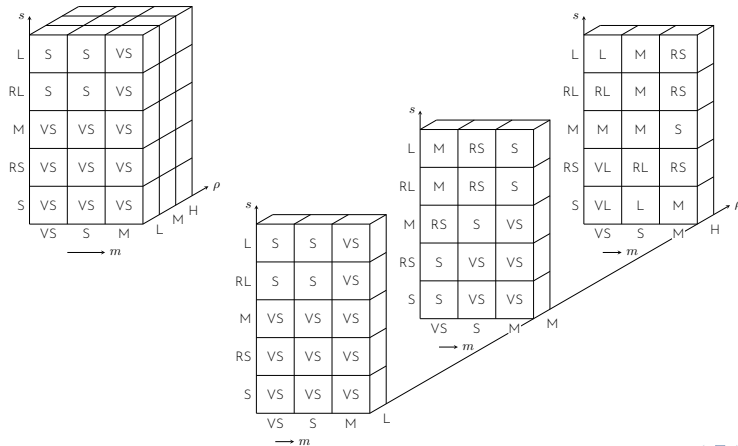
The rules are updated to include the two new fuzzy sets.



Case study 6

Step 5 (continued)

The rules are updated to include the two new fuzzy sets.

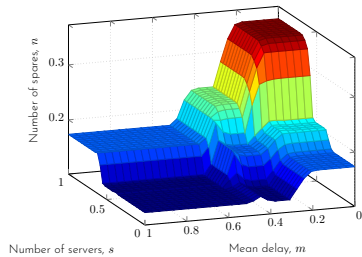


Case study 6

Step 5 (continued)

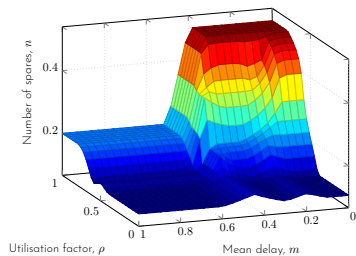
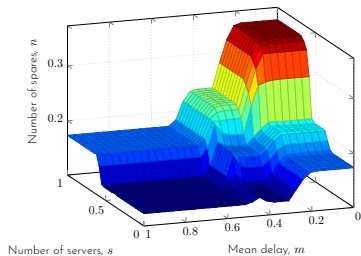
Case study 6

Step 5 (continued)



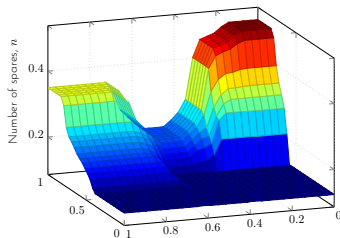
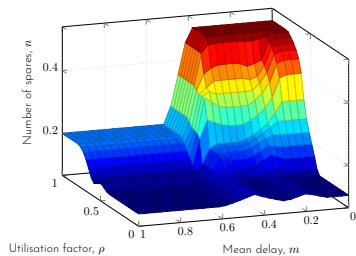
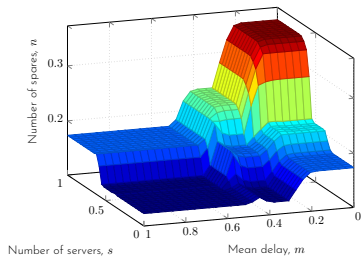
Case study 6

Step 5 (continued)



Case study 6

Step 5 (continued)



SUNWAY
UNIVERSITY

Jeffrey Cheah
Foundation

Nurturing the Seeds of Wisdom

- ▶ temperature of the room,
- ▶ number of people in the room,
- ▶ ambient temperature (temperature outside the room), and
- ▶ time to spend in the room,

Perform a inference for temperature of 30°C, 2pm, 20 people in the room, ambient temperature of 33°C, and to spend 1 hour in the room.

Example 8

How would it be different if you were to construct a Mamdani-style fuzzy system?

CSC3034 Current Course Update

03 Fuzzy Inference and Fuzzy Systems - presentation

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current Lab 2 uses scikit-fuzzy to build and evaluate Mamdani control systems.
- The physical-AI extension applies fuzzy brake/throttle logic to a simulated robot.
- The updated solution checks optional dependencies and supports a standalone fallback.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.